

**The FLAVIA® Creation 600
TECHNICAL MANUAL**

Part No: DOC-000390 Rev. 1



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Date	Issue	Section	Modifications and/or additions	Signature
May 2021	1	All	First Issue	

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General Information

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1.1 Overview

Introduction

This service manual covers the current models of the FLAVIA® Creation 600. These models can be distinguished from previous FLAVIA® models by their appearance and serial numbers.

Serial number labels are located in various locations including under the rubber mat on the top of the brewer and contain two codes, the profile number and the machine serial number, in the following format:

<Profile number> <Machine serial number>

For example: **SAB1NA 00002118**

The **Profile Number** is made up of the following:

SA<Machine Version><Machine Revision><Country Variant >

Where **SA** denotes the FLAVIA® C600.

<Machine Version> is one of:

- **B** = Brewer specification variant

<Machine Revision> is one of:

- **1** = First revision of the above

<Country variant> is one of:

- **JP** (Japan)
- **NA** (North America)
- **UK** (United Kingdom)

The **Machine Serial Number** is a unique incrementing sequential number, starting at '00000001', that is **not** zeroed if the Machine Version or Country variant changes. For example:

SAB1NA 00035657 would proceed **SAB1NA 00035658**

This manual is designed to act as a ready reference for Service Engineers who have received formal technical training in these models.

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1.2 Safe Working Practices

Electrical Safety

- ☒ **Do** isolate the machine from the electrical supply before attempting any repair where there is any exposed connector at a potential of more than 50 V AC.
- ☒ **Do** verify that the machine is electrically safe by suitable testing after completion of a repair.
- ☒ **Do not** under any circumstances, modify or bypass safety circuits.
- ☒ **Do not** attempt to repair conductors in any heater element circuit. Replace the affected loom complete.
- ☒ **Do not** leave unattended any machine that is switched ON with its external covers removed.

Earthing Installation

DANGER Electric Shock Hazard



Touching some of the internal components can cause serious personal injury or death. Do not disassemble this appliance without undertaking official training from Lavazza Professional.

WARNING

Electric Shock Hazard

Improper earthing can result in electric shock.

This appliance must be earthed. In the event of an electrical short circuit, grounding reduces the risk of electric shock by providing a current path for the fault current. This appliance is fitted with an earthed plug and power cord. The plug must be plugged into an outlet that is earthed and has the correct Live/Neutral polarity.

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1.2 Safe Working Practices

Service engineers should consult a qualified electrician if any doubt exists as to whether the electrical supply outlet is correctly installed.

1. A power-supply cord is provided to reduce the risks resulting from becoming entangled in or tripping over a longer cord. This cord is 1.75m for NA brewers and 1.95m for UK machines.
2. Lavazza Professional advises against the use of extension leads as they could present the risk of a trip or fire hazard.

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1.2 Safe Working Practices

General Safety

- ☒ **Do not** work on a machine without the client's knowledge.
- ☒ **Do not** obstruct passageways.
- ☒ **Do** clear up any water spillage as soon as they occur.
- ☒ **Do** familiarise yourself and comply with the client's own health and safety requirements.
- ☒ **Do** clear away any packaging, litter, etc, when the work is completed. Observe any recycling requirements in force.

Hygiene

- ☒ **Do not** allow water or food contact components to become soiled.
- ☒ **Do not** use proprietary cleaning agents on or near water or food contact components.
- ☒ **Do not** smoke while in the vicinity of the machine.
- ☒ **Do** maintain the highest standards of personal hygiene.
- ☒ **Do** wash your hands thoroughly prior to any work on the machine.
- ☒ **Do** leave the machine clean when you have finished.
- ☒ **Do** use only Lavazza Professional genuine replacement components to prevent possible tainting of the drinks.

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1.3 Machine Specification

Machine Specification

Weight (kg)		14.5kg / 32lbs	
Height (mm/inches)		435 mm/17.1"	
Width (mm)		308 mm/12.1"	
Depth (mm)		510 mm/20.1	
Power Supply (V AC)		120V - 60 Hz (NA)	230V - 50Hz (UK)
Typical Current (A)		13A (NA)	6.75 A (UK)
Rated Power Input (W)		1550 W	
Nominal water heater temperature (deg C)		92	
Typical drink temperature in paper cup (deg C)		84	
Recommended Drink Volumes (ml) with pack	NA	Reg. Tea / Coffee:	207
		Espresso:	89
		Tea:	207
		Choco:	177
		Cappuccino:	207 (excl. foam)
Recommended Drink Volumes (ml) with pack	UK	Reg. Coffee:	180
		Espresso:	60
		Tea:	160
		Choco:	155
		Cappuccino:	210 (excl. foam)
Cycle Time (sec)		Standard:	49
		2-pack:	89
Water supply requirements		Cold potable sodium free water via External drinking water carbonate hardness filter (required)	
		Minimum Flow Rate	2.0 L/min (0.53 US gal/min)
		Dynamic water pressure	Min. 1 bar/ 15 psi
			Max. 6.8 bar /100 psi

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1.4 Software Revision History

The history of production release versions of software for each of the two control systems in the FLAVIA® C600 is as follows:

Firmware Version	Date	Software Version	Date	Description
1.09	13/02/21	1.9.2.0	17/02/21	Current at time of document release

The availability of new Software updates can be found in:
Engineering menu->Technician Settings->Flash/Clone
or by going through installation mode:
Settings->1441->Technician Settings->Operating Modes->Installation
and letting the brewer check for updates.

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Installation

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2.7	Optional Modules

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2.1 Client Information

Checklist

- Intended machine location.
- Place on a stable surface and place out of the reach of children.
- Whereabouts of machine to be installed and product.
- Name of key operator and relief key operator (the persons responsible for cleaning and filling the machine).
- What level of drinks audit, if any, will be required?
- Availability of client and key operator(s) for demonstration and training.
- Any special circumstances, e.g. safety requirements.
- Advise the client of the time taken for installation and the key operator(s) training requirements.

Check that the following services are available:

- Potable water supply with between 1 and 6.8 bar pressure terminated within 0.8 m of the machine by an accessible stopcock.
- The stopcock should be terminated with the following connection: 15mm OD x $\frac{3}{4}$ BSP Male (WRAS approved or equivalent) isolation valve.
- Electrical supply mains terminated within 1.5 meters of the machine by an accessible switched socket with suitable current rating (values dependent on region. See Section 1.3 for details).

Services must be accessible without moving the machine.

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2.2 Overview

Safety compliance warning!

All electrical and plumbing directives in this section are mandatory in the UK only, but are recommended as good practice in non-UK countries. If in any doubt please follow local guidelines.

Overview

- 1) Check for transit damage.
- 2) Remove the packaging materials from the machine.
- 3) Ensure the machine is fitted to a stable, level surface
- 4) Do not install where a water jet could be used.

Electrical Supply

You will need to connect to a suitable switched electrical supply to meet the requirements for isolation of electrical equipment. (Refer to rating label on rear of machine or user guide.)

Do not connect any other appliances to the socket that your C600 is connected to.

Your brewer comes with a mains lead and has a ground wire with an earth plug. Only this lead should be used - no old leads should be re-used.

The plug must be plugged into an outlet that is properly installed and earthed.

Plumbed Systems

A plumbed brewer will need mains drinking water supply.

A stopcock should be terminated with the following connection:
15mm OD x 3/4" BSP Male (WRAS approved or equivalent) isolation valve.

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2.2 Overview

An in-line taste, odour, and hardness filter must be fitted on the mains water supply to this machine (not supplied) in order to achieve the best possible taste and quality from our drinks. This will also avoid premature failure due to scale related deposits.

Water pressure must be between 6.8 bar (100 psi / 0.68 MPa) and 1 bar (14.5 psi / 0.1 MPa).

Brewer to be connected to a potable cold water supply only. Chemically softened water is not suitable for use with this product.

NOTICE:

- Do not use water that has come from a water softener or contains sodium.
- Install in accordance with local plumbing regulations.
- Jointing compound should not be used in the water supply to the system, as this will affect drink flavour, only use PTFE tape if required.
- Do not locate the water supply above the mains power socket.
- Any questions, please consult a qualified electrician or plumber.
- We draw your attention to the requirements of the Water Supply (Water Fittings) Regulations 1999.

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2.3 Tools Required for Installation

Recommended tools

Torque spanner @ 3 Nm with 32mm open-ended attachment

John Guest ring attachment

Adjustable Spanner 10 Inch

10mm ring spanner

Adjustable Water Pump pliers

Phillips No.1 screwdriver

Phillips No.2 screwdriver

RCD Adapter

Safe Knife

Mains Polarity Tester

Paper towels

5-litre Bucket

¼ inch Plastic Tubing Cutter

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2.4 Installation Sequence

This sequence covers the installation of the FLAVIA® Creation 600.

Installing Plumbed-in Machines

- Locate hose and water blocker. If replacing an existing machine, be sure to use a new hose and water blocker.



- Confirm water blocker is set at mark 2. This will prevent an excess of 10 liters of water flooding the premises.
- Ensure that the hose is new and not re-used.

- Connect water blocker to water supply via stopcock, fit 15mm to ¼" BSP brass fitting found in the packing kit if necessary.



- Check clients supply is safe, secure and not leaking before making any connections.
- If it is necessary to use the brass fitting, ensure a new compression olive is fitted as reused ones can cause leaks.

- Tighten water blocker with 32mm open-ended torque spanner if available.

- Ensure spanner is calibrated and set to 3Nm

- Connect the straight end of the hose to the water blocker and torque with John Guest attachment.

- Ensure there is an in-line water filter (if not, please install an in-line water filter)

- Turn on mains water supply and flush approximately 25 litres of water through the customer's pipe work into waste bucket until the water runs clear.

- Ensure water runs clear to prevent contaminating the machine tanks.



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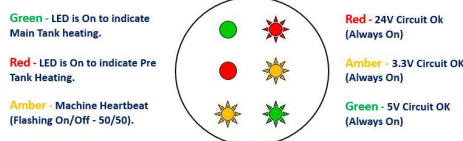
2.4 Installation Sequence

- Connect the hose to the machine inlet valve; tighten with John Guest attachment and torque spanner.
- Turn on the water supply and check water blocker and inlet valve connections for leaks.
- Test mains electrical supply.
- Plug RCD into wall socket. Test RCD as per the manufacturer's instructions. If RCD passes test, plug the machine's mains plug into RCD.
- Switch the brewer on using the ON/OFF switch on the back-left side of the brewer.
- Please follow the on-screen prompts and enter the following information:
 - a. Select Language
 - b. Select Country
 - c. Ensure the customer reads and accepts the End-User License Agreement
 - d. Enter Location Information
 - e. General Settings-
 - i. Water Source- two options available, "plumbed-in" or FLAVIA® REFILL. FLAVIA® REFILL is an optional external water tank that allows the C600 to be jug filled. Please refer to section 2.7 for more information on the FLAVIA ® REFILL.
 - ii. Max Hot Volume- This can be set from 5oz to 9oz

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2.4 Installation Sequence

- iii. (147ml to 266ml) depending on the max cup size at the client's site.
 - iv. Pack Bin- two options available, Internal and External. When using an External bin funnel (Part #76-24-0901) ensure that the worktop has an appropriate sized cutout (ref. 70-40-0680 – Footprint layout C600) with an appropriate waste bin positioned underneath. Choose external pack bin to disable pack bin counter. When choosing "external" the brewer will no longer prompt users to empty the pack bin.
 - v. Energy Saving- This comes default set to 4 hours but can be turned off or adjusted to fit your client's needs. Please refer to section 7.1 for more information.
 - f. Drinks and Packs- The drinks and packs section set what drinks will be available on the drinks carousel for the customer to choose from. Select drinks and packs based off the customers Freshpack Merchandiser.
 - g. Time and Date
- At the end of the installation menu, the brewer will attempt to connect to the internet and download the latest software revision.
 - The brewer will ask if the water line has been connected and shows an interactive video. Press the "Connected" button and the flushing cycle will start. This process runs water through the pump, both hot tanks (pre-tank and main tank), and the dispensing water system.
 - Check the Brewer's status LEDs. These can be viewed through the circular window in the left-hand side of the brewer chassis once the pack bin has been removed. Flashing red and green LEDs show the tank heating control during the water heating cycle.



Note: only one heater will be energised at any one time, with the Main tank having priority over the Pre tank.

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2.4 Installation Sequence

- **Notes** about priming:
 - a. You cannot make drinks if you have not first primed the hot tanks.
 - b. At the completion of the priming process, if the water level in the cold tank has dropped below the float switch then the heating-up will be delayed until the cold tank refills.
- If a PCO error occurs during the priming process then this will abort that process and the menu status will revert to the initial 'Disabled (Tanks Empty)'.
- **Flushing the Brewer**

Once the brewer is primed, the screen will prompt you to flush the brewer. Follow the interactive video on screen that will include the following steps:

 - a. Remove the drip tray
 - b. Partly slide out the Freshpack bin far enough to capture the water, which will be dispensed as illustrated to the right. Note that the flushing cycle is intermittent to allow time for the cold tank to refill and will pause and restart automatically whilst the screen displays "Tanks Flushing". The flush cycle will automatically stop once it is complete.
 - c. After the flush, empty any water from the Freshpack bin, clean the bin, and fully install it back into the brewer.
 - d. The brewer will require 2 minutes to heat the water.



Please note: Flushing the brewer can also be done at anytime from the Operator Menu. **Please refer to section 7.1 for more information and how to access Operator Menu.**

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2.4 Installation Sequence

It is recommended that a flush procedure be performed if the brewer has been in storage, shut down for more than 5 days, or the water system has been worked on.

Once the priming/heating cycle and flush is completed, the brewer will show you the main screen drink carousel (please see example image below).



Drinks Carousel Example

Please note that all the brewer's installation settings can also be found in the Operator Menu/Settings. Please refer to section 7.1 for more information and how to access Operator Menu.

- Vend a coffee by either touching the coffee selection or clicking the “Touch to Insert Pack” button. The C600 is equipped with a bar code scanner and can identify what type of pack is inserted. This allows the client to shop via the drink's carousel or by the Freshpack Merchandiser.
- Vend a minimum of 5 filter coffees to calibrate the drink volumes (no manual adjustment should be required), and then check that the drink volumes are within specification (refer to Section 1.3).
- Re-check all the leak points (Shut-off valve, water line, and inlet valve adapter).
- Setup the Merchandiser; label with packs and fill.

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2.4 Installation Sequence

- For any other configuration settings, refer to Sections 7.1 and 7.2.
- Vend a weak tea and check the drink for taint.
 - 
 - Confirm that NO bad tastes are present
 - If any bad tastes are detected, drain the cold tank, refill and flush hot tank using Hot Water button.
- Access the 'Pack Heater:' menu in Engineering Mode and turn this ON. The Pack fan will also run automatically for 30 secs. Repeat this one more time to burn off potential dust and smell on the Pack Heater.

Note:

This may also occur if the machine has been in storage for a length of time.

If the smell lasts for more than a few vends or if you have **any doubts whatsoever** then disconnect the machine immediately and contact your supplier.
- Disconnect the RCD adapter from the mains power outlet and re-connect the mains plug into the mains supply.
- Fit a Sales Service enquiry sticker adjacent to serial number near the pack bin area.

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2.5 Key Operator Training

The FLAVIA® C600 can prepare Filter Coffees, Teas and Chocolates as well as specialty drinks like Frothy Cappuccinos, Lattes and Mochas. Milk based beverages can be created using a user's own milk, their own milk substitute or by using a Fresh-pack such as Real Milk Froth. When a milk-based beverage is selected, the brewer will prompt for the relevant actions and open the pack door. Note that this functionality is configurable via the "Drinks and Packs" section within the operator menus (See Section 7.1)

Training steps

Train the key operators in the operation of the machine, using the User Guide as a script. It is in the installer's interests to ensure this training is effective, otherwise preventable machine breakdowns may occur.

1. Introduce key operator to C600 and demonstrate how to isolate power and water. Give an overview of the machine, i.e. the touchscreen, the electronic cup stand and cup sensors, the barcode scanner and pack recognition, etc.
2. Point out important elements of the brewer so that the customer understands the proper names and functions of these parts.
3. Using the User Guide, go through the "Daily Care" and "Weekly Care" (pages 24-30)



- Allow customer to demonstrate back to you, to check their understanding.

Demonstrate...

- a. How to prepare and vend a drink. Include information about pack recognition with the bar code scanner.
- b. Drink selection via the drinks carousel or by inserting a pack.
- c. The automatic cup-stand and the removal of the cup-stand for larger travel mugs (please refer to the User Guide).
- d. Pack bin removal and how and why it may show "please empty the pack bin" message.
- e. Pack bin "pack counting" and why it is a good practice to always empty the bin if it is removed. If the bin is not emptied, then the customer should select "No" when the machine prompts "Did you empty the pack bin before replacing it"

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2.5 Key Operator Training

- f. Pack door removal, cleaning, and re-fitting.
- g. Ensure air duct is clean and free of debris.
- h. Left and right cup sensor rails and the importance of cleaning them.
- i. Cleaning the bar code scanner lens.
- j. Cleaning the drip-tray and the proper assembly of the cup stand.
- k. Draining the cold-water tank.
- l. Stocking the merchandiser and changing selection labels.

Configure Energy Saving Mode

In order to conserve energy, the brewer can be configured to switch to an energy saving standby mode after a specific time of inactivity. To configure the energy saving mode:

1. Enter Operator menu by holding your finger in the top left corner of the screen for about 5 seconds.
2. Enter password 3-3-3-3 and touch “ok”.
3. Select Site Preferences and then Energy Saving.
4. Select the time period for the brewer to enter energy saving mode.
5. Touch the back button in the top left-hand corner 3 times to return to the drink’s carousel.

If the brewer is inactive for the specific time set, it will enter energy saving mode. In this mode the boilers will be switched off and the display will show a standby symbol. Touch anywhere on the screen and it will exit energy saving mode. The brewer screen will say “Please wait while water is heated” and in approximately 2 minutes the drinks carousel will appear.

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2.5 Key Operator Training

Set Operator Password

The operator menu of the brewer can be password protected to prevent unauthorized access. To set a new password:

1. Enter Operator menu by holding your finger in the top left corner of the screen for about 5 seconds.
2. Enter password 3-3-3 and hit "ok".
3. Click on Site Preferences and then Password.
4. Select the 4-digit password that you would like to replace the default with.
5. Press the back button in the top left-hand corner 3 times to return to the drink's carousel.

Note:



If a password is lost or forgotten, only a service engineer can reset it. Changing the Operator Menu password is not recommended.

All machines:

1. Wipe clean the machine exterior.
2. Ensure the machine is left to Customer's requirements.
3. Ensure the key operator is aware of the troubleshooting section of the guide and instruct in the repairs possible by the key operator.

Demonstration

If possible, demonstrate to the client that the machine installation is complete and that the key operator(s) has been trained. Confirm that they are satisfied.

Demonstrate machine operation to general users if possible to 'get people using the machine'.

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2.5 Key Operator Training

Notes on this section

- Some differences in menus may be apparent, depending on the version of software fitted to the machine, and configuration selections made within the menus.
- If in any doubt, contact Lavazza Professional technical help desk for further information.
- More detailed information of the menu modes available on the machine is contained in Section 7 of this manual.

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2.6 Moving the Brewer

Note: Do not move a brewer when the cold tank is full of water!

Drain, Disconnect and Move

1. Turn the brewer off using the ON/OFF switch on the back bottom left-hand side.
2. Unplug the brewer from the electrical outlet.
3. Follow the water line in the back of the brewer until you find the shut-off valve and turn the valve to off.
4. Remove the water inlet valve adapter from the back of the brewer (adjustable water pump pliers may be needed to loosen the adapter).
5. Remove the cup stand and pack bin in order to access the cold tank drain on the right-hand side (cover marked in red). Please see the example picture below:



6. Remove the red cover and pull the cover towards you. The cold tank drain tube is attached to this. Drain the cold tank completely. The empty pack bin can be used as a draining bucket.
7. Replace the cold tank drain tube cap and clip the cover back into its original location.

If the brewer is being moved to a new site, the hot tanks must also be drained using the following procedure.

8. Remove the right hand cosmetic and safety cover.
9. Unclip the hot tank drain hose from the chassis clinch feature.
10. Remove the red plug from the hot tank drain hose

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2.6 Moving the Brewer

11. Replace the right-hand safety cover, feeding the drain hose through the cut out in the bottom rear of the cover.
12. Re-power the brewer.
13. Drain the hot tanks by accessing **Engineering Menus**.
 - a. Enter Engineering Menu by holding your finger in the top left corner of the screen for about 5 seconds.
 - b. Enter password 1-4-4-1 and press “ok”.
 - c. Click on Engineering Menus.
 - d. Ensure the hot tank drain hose is over a suitable receptacle for hot water.

CAUTION – the drained water and drain hose will be hot
 - e. Click the back button to “Hot Tank Drain” and touch “Drain” to empty the hot tanks. The hot water will be driven from the system using the air pump located near the manifold. Once the tanks have been drained the tanks will be disabled until the system is re-primed in its new location.
14. Enter the Technician Menu (1-4-4-1) and change the Operating Mode to **Installation**
15. Power down the brewer.
16. Replace the plug in the hot tank drain hose and secure it with a cable tie.
17. Re-insert the drain hose in the chassis clinch feature (See Section 6.5.4. for details),
18. Replace the safety side-cover.
19. The brewer may now be moved safely. You may find it convenient to store the drip-tray inside the pack bin during transit. Removing the magnetic side covers will expose convenient lifting handles.

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2.7 Optional Modules

The FLAVIA® Creation 600™ has two optional attachment modules, The FLAVIA® CHILL™ Cold module and the FLAVIA® REFILL™ water tank module. These modules were created to enhance the usability and functionality of the FLAVIA® Creation 600™.



2.7.1. FLAVIA® CHILL™ (NA region only)

The FLAVIA® CHILL™ is a cold module attachment to the C600. It was designed to add options and versatility to the C600 giving customers the ability to brew both hot and cold single serve beverages from one easy to use platform. The CHILL™ allows customers to brew the following beverages:

- Cold Brew Coffee
- Iced Coffee
- Iced Teas
- Infused Waters

Installation of the FLAVIA® CHILL™ must be done by a trained distributor technician/installer. Installation and safety instructions can be found in the

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2.7 Optional Modules

FLAVIA® CHILL™ Technical/Service Guide. Setup and care of the FLAVIA® CHILL™ can be found in the FLAVIA® CHILL™ User Guide (70-50-0153).

2.7.2. FLAVIA® REFILL™

The FLAVIA® REFILL™ is a replacement option for a connected water supply. The water tank with a capacity of 213.0 fl oz (6.3 L) with its water filter installed acts as a water reservoir for the drinks that the connected brewer produces. Therefore, the brewer will not need to be connected to a water supply. This setup is a great solution if there is a need to locate the brewer in a position far away from a water supply. REFILL supplies water directly into the brewer's internal water tank (not via the brewer's 3/4" inlet connection).

This must be connected to the brewer by a distributor technician.

Installation, safety, and care instructions can be found in the FLAVIA® REFILL™ User Guide (70-50-1061). Please note, the REFILL™ was designed to attach to the left or right side of C600.

2.7.3. Enabling Optional Modules

The following instruction are required to configure the C600 once an option module has been installed.

1. To select and enable your option C600 Module:
 - a. Enter Technician Menu by holding your finger in the top left corner of the screen for about 5 seconds.
 - b. Enter password 1-4-4-1 and press "ok".
 - c. Click on Modules.
 - d. Click on the module that you have attached to C600 (either Chiller or External water tank).
 - e. Change "not fitted" to "fitted" on the module that you have attached to C600.
 - f. Back out of the menus using the back arrow in the top left-corner of the screen.

Please Note, when enabling CHILL™ please reference CHILL™ User

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2.7 Optional Modules

Guide page 15 “Adding Cold Drinks to the Main Brewer Menu” for instructions on adding the cold drinks to the drink’s carousel.

2.7.4. Other Optional accessories

For full details of all spares, consumables and optional accessories see C600 Spares list (76-00-0442), but a subset of the most commonly used are shown below.

2.7.4.1. CHILL 1.1 upgrade kit

If a site has a C600 and CHILL module, this kit allows the CHILL module to be positioned remotely from the brewer.

2.7.4.2. Aerial Externalization kit (76-10-5575).

If the C600 is located in an area of poor cell reception that prevents connection to IOT, this kit provides an external aerial that can be located in an area of better reception. Installation should be completed by a trained technician in-line with the instructions details in Section 6.6.4

2.7.4.3. RJ45 adaptor kit (76-10-5567).

If a hardwired connection is preferred for the IoT connection then this kit can be used to provide a female RJ45 connection on the rear of the brewer which will accept an ethernet cable (not provided). Installation should be completed by a trained technician in-line with the instructions details in Section 6.6.4.

2.7.4.4. Bin Funnel Assembly (76-24-0901)

If it is possible to cut a hole in the worktop, then the bin funnel allows for spent fresh-packs to be diverted in to a large under counter bin (not supplied). Worktop cut-outs should be made in line with 70-40-0680 (footprint layout C600)

2.7.4.5. C600 Particle screen and activated carbon filter (76-24-0634)

It is recommended to replace this every 24 months.

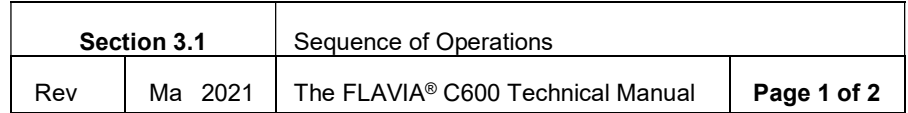
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Sequence of Operations

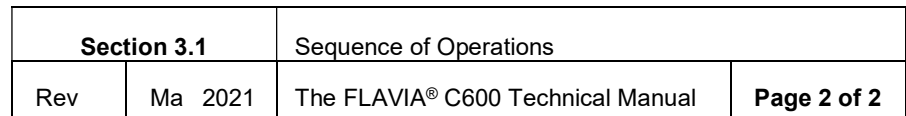
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- 3.1** Water System Schematic
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3.2 Sequence of Operations

Not Currently Available

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Installation

Contents

- 4.1** Overview
- 4.2** System Schematic

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4.1 Overview

This section covers the various electrical circuits of the C600 brewer. It should be read in conjunction with the C600 system wiring diagram (70-08-1836) and the circuit diagrams for individual circuit boards within the brewer.

The brewer is controlled by a software system that is physically distributed over two processors. One located on a circuit board attached to the head unit (SBC) and one located on the large circuit board to the side of the brewer (BrewCon). Communications between the two software systems is achieved via a serial link that consists of two cables. See pg1 of the system wiring diagram

The software system reads data from various sensor inputs and outputs instructions to mechanical actuators as needed to vend a drink. This section of the document groups inputs and outputs together to form functional subsystems of the brewer.

The functional subsystems are...

- User interface
- Water system
- Cup control
- Pack heater
- Pack handling
- External module interfaces
- External communications

4.1.1. User interface

This functional sub-system handles Interaction with a user via a touch-sensitive LCD display screen. It consists of software running on the SBC PCBA, A touch sensitive colour LCD display, an interface PCBA and some interconnecting cables. See page 1 of the system wiring diagram

The software sends image data to, and receives touch information from, the screen via the interface PCBA. This PCBA also creates the necessary voltages to power the display panel and its backlight.

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4.1 Overview

4.1.2. Water system

This functional sub-system contains the elements necessary to control the water system within the brewer. There is a sub-system to maintain the fill level of the cold-water tank, a sub-system to heat the water and a sub-system to control the flow of water into the cup. All three sub-systems are controlled by software running on the BrewCon PCBA

Cold tank control:

The fill level of the cold-water tank is controlled by a float switch located inside the cold tank (SW7) to ensure there is sufficient water to complete a vend. The system consists of software, running on the BrewCon PCBA, monitoring the state of a float switch in the tank and activating solenoid valves, mounted on the rear panel, to allow water into the cold tank. See page 5 of the system wiring diagram.

Water heater control:

Water is heated and stored within the brewer in two hot tanks that are electrically heated to maintain a target temperature of 94 degrees C. The system consists of software, running on the BrewCon PCBA, monitoring the temperature of the water in each tank. It heats the water by activating a mains powered heating element in each tank. In addition to software control, each tank also includes two levels of thermal cut-out, a resettable fuse ($229 \pm 0/-10$ °C) and Wax fuse (105 ± 5 °C), to ensure the water cannot be overheated. See pages 4 and 7 of the system wiring diagram.

Water delivery control:

Water is pumped from the cold tank into the hot tanks which forces hot water out into a distribution manifold. Three valves on the distribution manifold can direct water to either the injector nozzle, the jet nozzle or back into the cold tank.

The system consists of software, running on the BrewCon PCBA, which monitors the water flow and manifold pressure for each vend. It controls an electric water pump to maintain the necessary flow rates needed during a vend. An over-pressure switch is used to protect the product packs from being over-inflated and pauses if the pressure in the system gets too high (9.5 ± 1.0 PSI). Following a vend, water is forced from the dispense pipework using an electric air pump. See page 4 of the system wiring diagram

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4.1 Overview

4.1.3. Cup control.

This functional sub-system contains the elements necessary to ensure the user's cup is positioned at the correct height to prevent the vend from creating mess. The cup-stand automatically detects the presence of a cup and raises it until the rim is at the correct height for drink delivery. It is capable of operation with both opaque and most transparent cups.

The sub-system consists of software, running on the BrewCon PCBA, using light beams to monitor both the presence of a cup on the stand and when the cup has been raised to the correct height. Three independent light beams cross the cup stand area at different heights. The light beams are very accurate and are sensitive enough to detect the rim of most transparent cups. The software also detects that the cup stand has been placed, by monitoring the state of the drip-tray detect switch, and it controls the movement of the cup using a motor fitted with a movement encoder. See page 3 of the system wiring diagram

The three light beams are created by a transmitting PCBA on one side of the cup area and a receiving PCBA on the other. The LED's on the transmitting PCBA are driven to maintain a precise level of light energy delivered to the detectors on the receiving PCBA. Anything that reduces the light energy will result in the LED's being driven harder to make up the difference. The software can compensate for slow degradation in the level of light reaching the detectors but cannot automatically compensate for sudden change like vend mess on the detector / LED lenses.

4.1.4. Pack heater

This functional sub-system contains the elements necessary to ensure the pack opens softly. This is to prevent vend mess. See pages 5 and 7 of the system wiring diagram.

The sub-system consists of software, running on the BrewCon PCBA, which is informed when a vend is underway and a pack is ready for heating. The software initially turns on the pack fan and confirms it is running via the returned fan speed signal. If all is ok, it turns on the mains powered pack heating element which heats the airflow being directed at the base of the pack. The heating period is timed by the software which turns

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4.1 Overview

off the heating element after 19 seconds but keeps the fan running to remove stored heat energy. The fan turns off after a further 30 seconds.

4.1.5. Pack handling

This functional sub-system contains the elements necessary to control movements of the pack from guiding its placement into the machine to its automatically disposed into the bin. See pages 6 and 8 of the system wiring diagram

Software running on the SBC PCBA detects that a user has made a drink selection and passes this information to the software running on the BrewCon PCBA. Initially the user is prompted where to place the pack by the opening of the pack door.

This is done using the door control motor. The software monitors movement of this motor using the door release cam switch (SW2). The door latch switch (SW3) in the motor assembly informs the software when the pack door has been closed and secured.

After the door is closed the software confirms a pack is correctly located, using the pack detect switch (SW6) in the handling mechanism. It also has the ability to check that the type of pack selected matches that inserted by using a bar-code reader located at the rear of the bin area.

When the pack is in place the software drives the injector into the pack using the DC mech motor. Movements of the motor and injector are monitored using two switches in the handling mechanism (SW4 & SW5).

Following completion of the vend the DC mech motor is used again to throw the used pack into the bin. The presence of the bin is confirmed by the status of the bin switch (SW1).

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4.1 Overview

4.1.6. External module interfaces

This functional sub-system contains the elements necessary for the brewer to communicate with up to three external modules directly connected to the rear panel of the brewer. There is an interface for a dedicated payment module interface, a dedicated water module and a general-purpose Pod module (Not currently available). See page 9 of the system wiring diagram.

The sub-system consists of software, running on both the SBC PCBA and the BrewCon PCBA. The software communicates with the pod via wire harness that have connections available on the rear panel.

External communications

This functional sub-system contains the elements necessary to allow the brewer to connect to remote devices using universally adopted communications channels. See page 8 of the system wiring diagram

The brewer can communicate using the following channels

- USB
- Wi-Fi
- Mobile telephone network
- Ethernet

This communication is handled by software running on the SBC PCBA.

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4.2 System Schematic

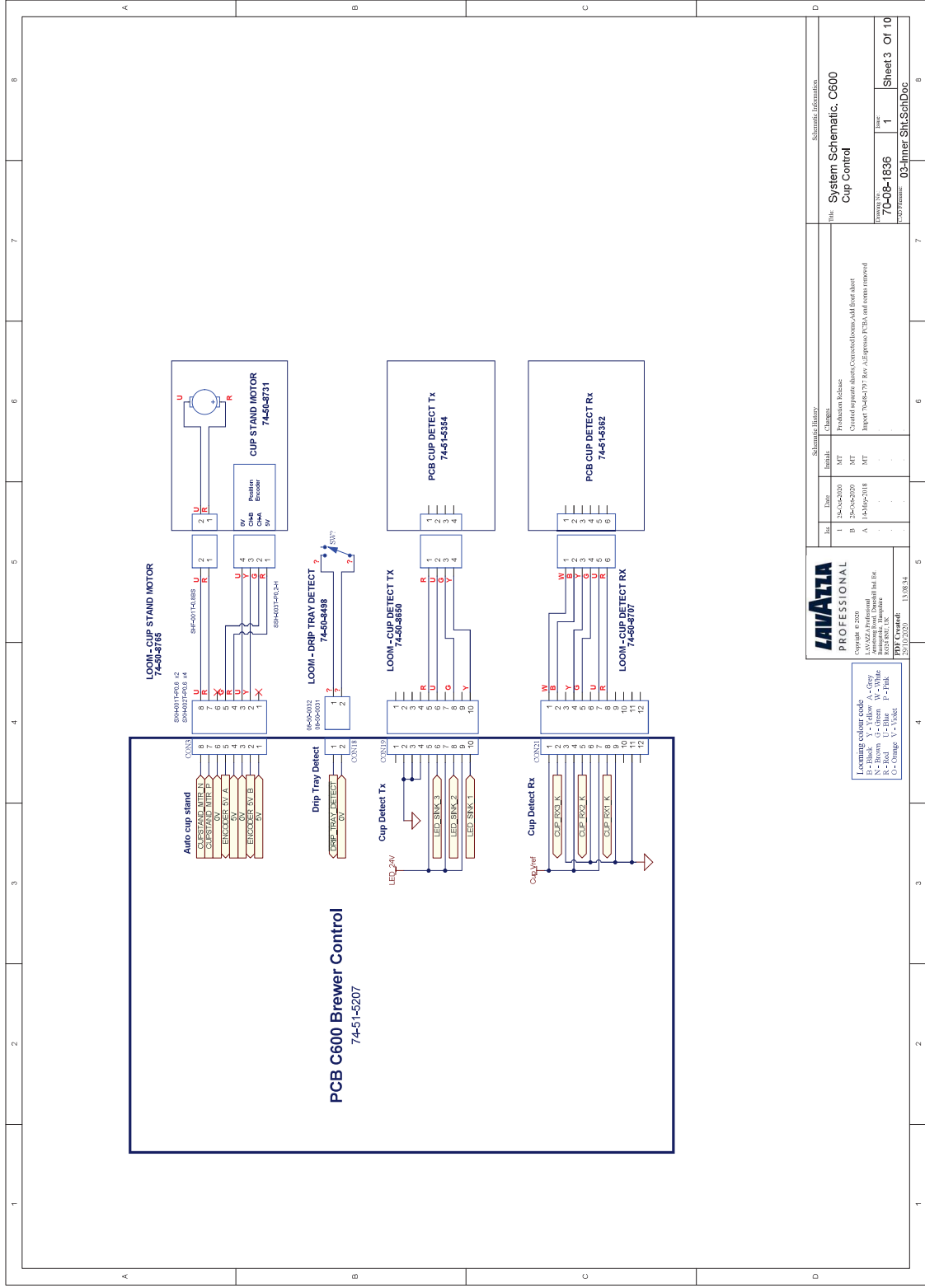
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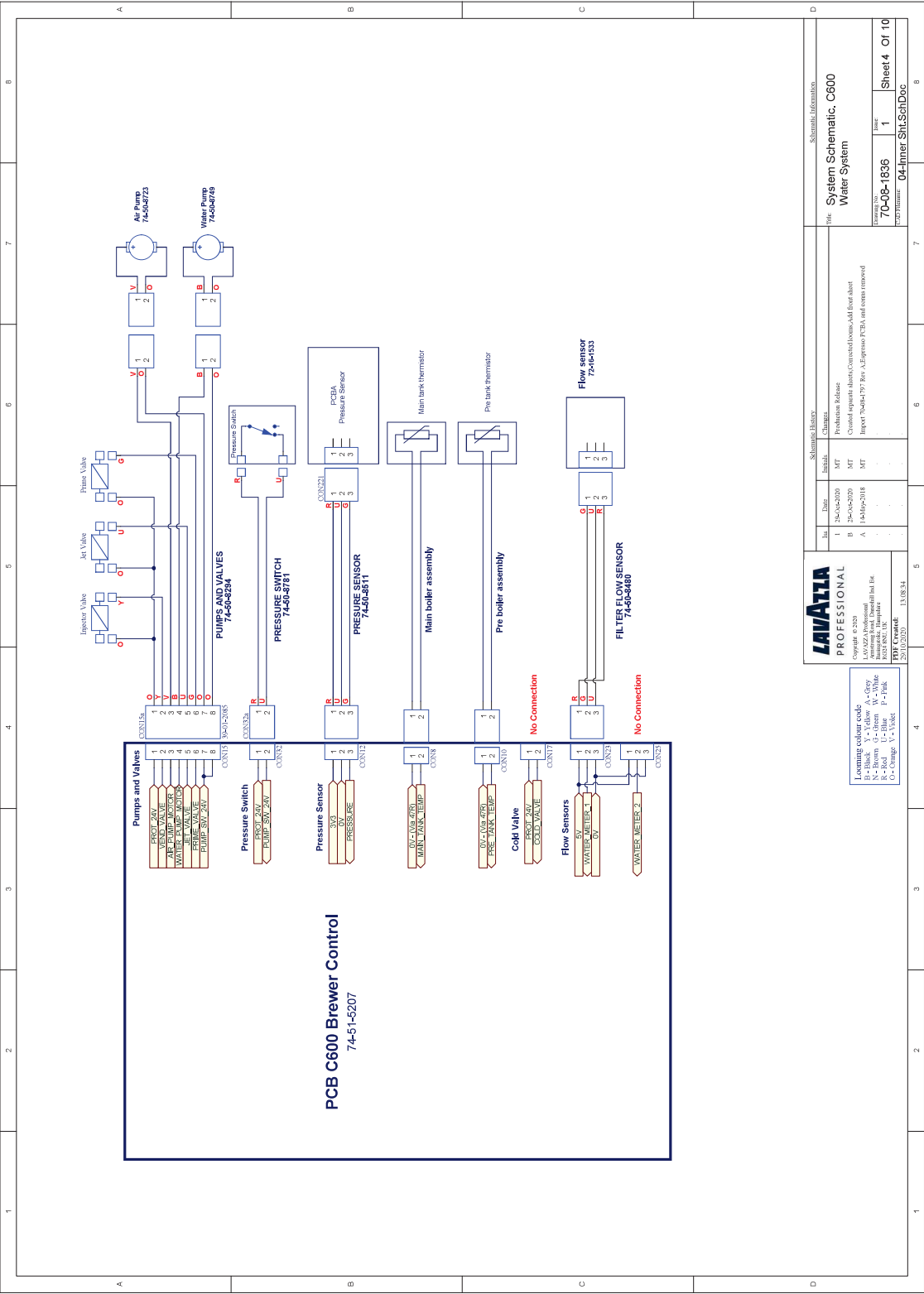
System Schematic. C600

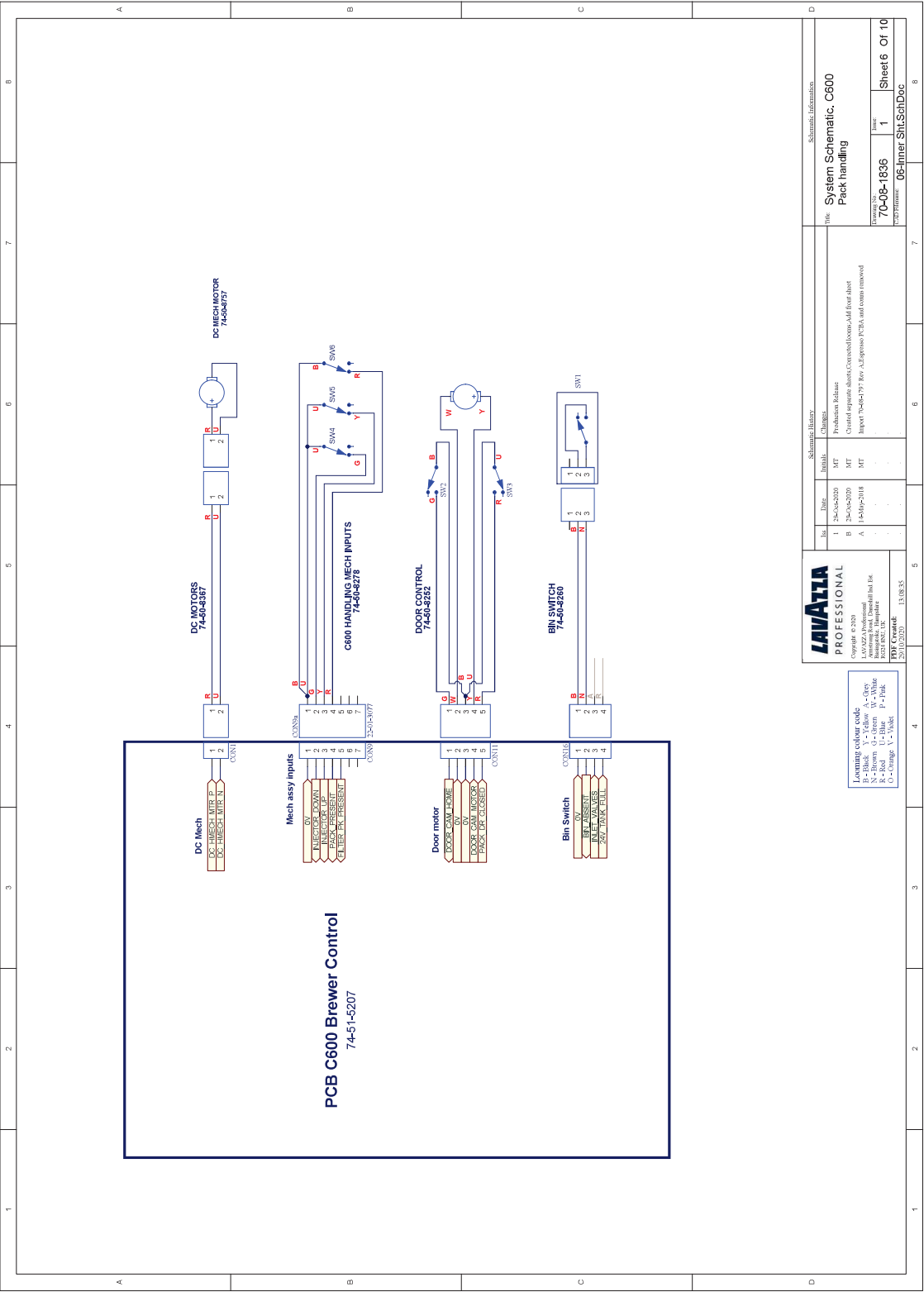
System Schematic

Drawing No: 70-08-1836 Revision: 1
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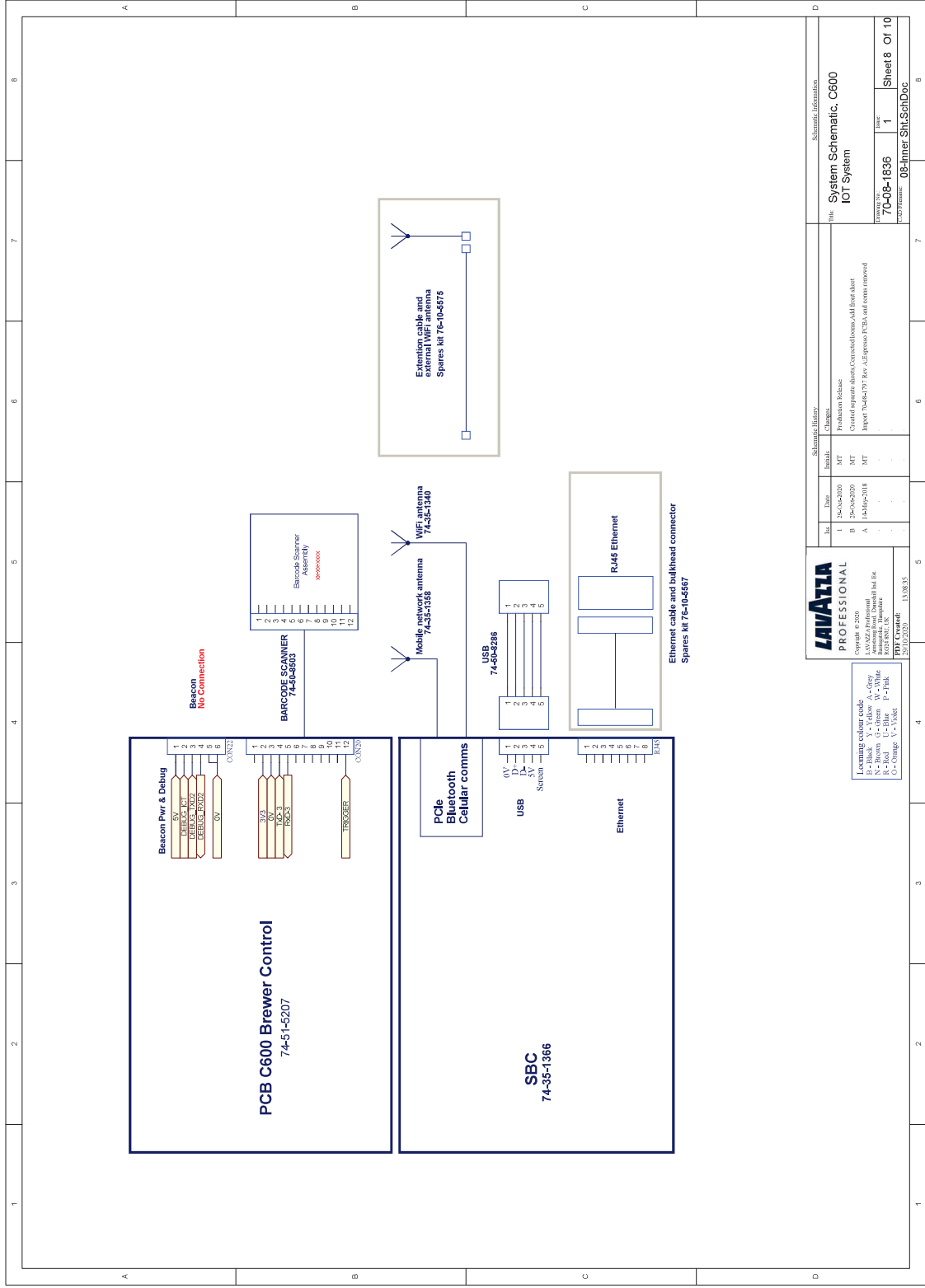
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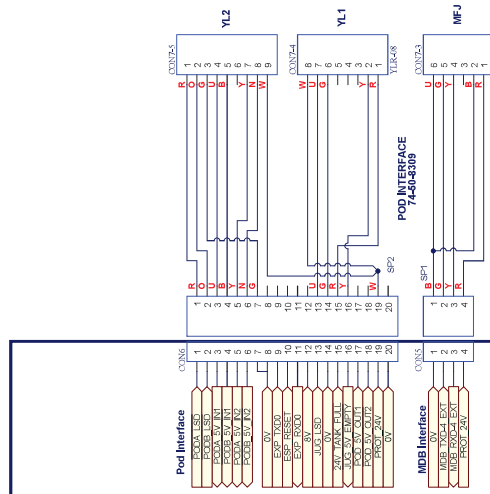
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System Schematic, C600
Pack handling

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Date: **1**
Sheet **6** of **10**

Drawn by: **06-Inner Sht.SchDoc**
Date: **8**





PCB C600 Brewer Control
74-51-5207

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Maintenance Recommendations

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5.4	Hot Water System
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5.6	Creation 600 Control Boards
5.7	Functional Test

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5.1 Overview

Introduction

This section contains some recommendations for preventative work to be carried out to maintain the reliability of the equipment at its designed level and to ensure client satisfaction when servicing various parts of the machine.

It is designed to be followed when an engineer attends a machine due to a breakdown or for a programme of preventative maintenance work.

If replacement parts or assemblies are required, they can be identified in the C600 Spares list (76-00-0442)

Critical Check Actions

1) **Description - Safety Critical**



This Critical Check symbol highlights the areas of the machine that are Safety, Electrical, and Hygiene critical.

We advise that actions be followed on each service call where this symbol is displayed. These checks are mandatory.

2) **Description - Reliability Critical**



This Critical Check symbol highlights the areas of the machine that are Reliability critical.

We advise that actions are followed on each service call where this symbol is displayed. These checks are recommended.

In this way, the client is made aware of the customer care you provide and a good many unnecessary calls are prevented.

Note: Preventative work that is recommended when servicing a specific area of the machine is described in the following sections.

Follow Lock Off Tag Out and Portable Appliance Test procedure at all times.

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5.2 Arrival

Arrival

Contact Customer to confirm arrival on site and if a BF ascertain fault and correct machine number/location matches the call

Check work area and machine is safe to work on.

Always report any unsafe or hazardous operational problems that you may encounter. Minimise the risks through your own awareness and conduct your own dynamic risk assessment every time you visit a site.

Follow Portable Appliance Test procedure at all times

	Action	Red Text = Machine On Black Text = Machine Off	Critical Check
1.	Anti-Bacterial wipe down – Clean all external surfaces of machine with antibacterial wipes.		▲
2.	Portable Appliance Test machine – Visually inspect mains cable integrity. Earth recommended pass <0.1 ohms. Insulation requires a pass of >2M ohms.		▲
3.	PCO & Event log – Note the number of PCO error occurrences from menu press top left of LCD then code [4-3-3-3].		■
4.	Maintenance history card - for filter change date and historical services (UK only)		■
5.	Water blocker hose and inlet valves - Check connections for leaks; repair or replace.		■
6.	Water Scale Filter – Ensure Scale filter is connected. Check for leaks and expiry date (change at 12 months max).		■
7.	Internal Particle Filter - Follow on screen instruction for flushing through in operator menu (3-3-3-3) change at 24 months max.		■
8.	Flavia Refill (if fitted) – Check for Leaks. Ensure filter is fitted and check for expiry date on history card. Change at a life of 1600 vends or 6 months max). Filter flush must be done after replacement. Ensure water filled to maximum level when entering priming and flushing program.		

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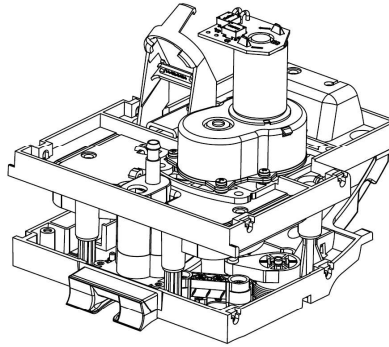
5.2 Arrival

9.	LCD touch screen - Check general condition, LCD display screen for evidence of poor contrast, condensation and that all configuration and selections are working.	
10	Vend-Look-Listen – • Cup stand motor – remove cup stand and replace. Motor will re-calibrate. Check for smooth operation. • Cup detect sensors • Bar code reader - test using Bar-Coded pack. • Touch screen • Pack opening • Cup volumes using a nozzle only. Vend a Hot Chocolate cycle: Target 163.5g (min 158.8 – max 168.2) • Drink temperature 78 - 89°c in a paper cup.	

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5.3 Pack Handling System

5.3.1. Handling Mechanism



The pack handling mechanism system is the heart of the machine and if properly maintained should give trouble free service.

Preventative maintenance

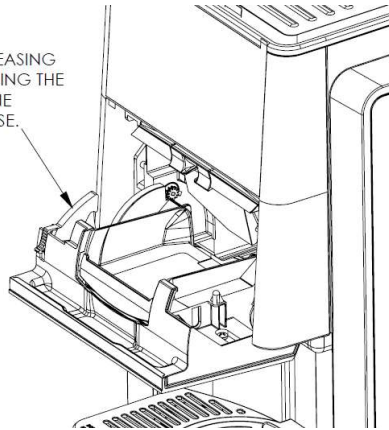
	Action	Critical Check
1.	Pack Rail - Check for damage, clean, and replace if necessary	■
2.	Pawl Arm - Check condition and operation. Ensure that arm travels past the back of the eject rail and is not fouling along its travel.	■
3.	Barrel Cam - Check tightness for backlash and wear, replace.	■
4.	Pack Injector - Inspect for scale, dirt build up, or blockages causing short shots or the pressure switch to trigger. Clean or replace injector if dirty or damaged.	■
5.	Handling Mech Motor - Refer to previous error log checks, for time outs. NB: If replacing either the assembly or the motor, calibration on inject & eject times MUST be conducted using "H Mech IJ Calibration" and "H Mech EJ Calibration" in the Engineering menu (See Section 5.6.6).	■

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5.3 Pack Handling System

5.3.2. Pack Door & Pack Barrier

REMOVE THE DOOR BY RELEASING THE ARROWED LATCH; PULLING THE DOOR OUT OF THE MACHINE PERPENDICULAR TO THE BASE.



The pack door mechanism is designed to last the life of the machine.

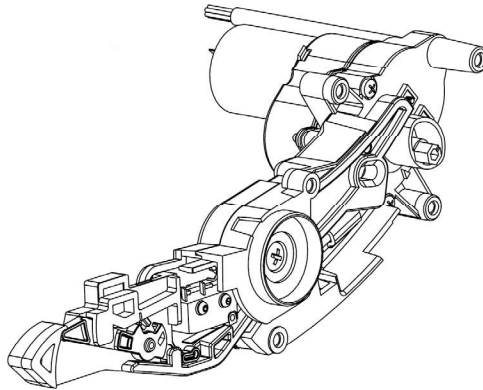
Preventative maintenance

	Action	Critical Check
1.	Pack Door - Ensure it opens smoothly. Clean internal and external surfaces using bacteria wipes. Apply food safe lubricant (72-31-0366) to both left & right hinges as shown in Section 6.3.1. The pack flap and hinge should be free moving.	■
2.	Pack Door Release Catch - Inspect for damage, the catch should move freely when pulled and return to its original position when released.	■
3.	Pack Barrier - The Pack Barrier should be free moving and return to the forward position when the door is open or removed. Check flap spring for corrosion, clean and lubricate hinge and spring.	■

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5.3 Pack Handling System

5.3.3. Door Latch



The pack door latch mechanism is designed to last the life of the machine.

Preventative maintenance

	Action	Critical Check
1.	Door latch release mechanism - Check that the latch assembly moves freely up and down when manually opening the door using the red release lever. With the door fully open, the latch should be able to move freely without friction, and spring return to its original position. Consider applying food safe lubricant (72-31-0366) if required.	■
2.	Door latch mechanism - Check that the door closes and that the latch operates smoothly and fully.	■

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5.4 Hot Water System

When servicing a machine, it is recommended that the cold water tank, float switch and associated silicone tubing and seals are also cleaned or replaced in order to prevent future problems. By flushing it out with clean water, you will remove any particles that have been introduced into the tank.

After re-making any water connection, it is advisable to check for leaks. Using the air pump to pressurize the water system and then using the pressure sensor readout to check for a fall in pressure after the air-pump has been turned off is a convenient way of detecting gross leaks in the majority of the water system, - Note that this cannot replace visual inspection or diligence and should be used as a convenient indicator only. Both air pump control and the pressure sensor readout are located in the engineering menu (See Section 7.2)

Lavazza Professional do not recommend the use of chemical descalents to remove lime scale on this machine.

IMPORTANT! Disable hot tank control in 'Engineering mode' prior to draining or attempting maintenance on the water system.

Preventative maintenance

	Action	Critical Check
1.	IMPORTANT! Disable hot tank control in 'Engineering mode' prior to draining or attempting maintenance on tanks.	▲
2.	Looms and connections - inspect all looms and electrical connection for integrity and signs of deterioration.	▲
3.	Hot tanks - check for leaks, tubing connection are secure and for any signs of discoloration or scale build-up.	▲
4.	Water pump motor - Replace if noisy or giving short measures. Ensure securing band is in place. NB: New pump calibration cycle must be run if replacing pump (See Section 5.6.6).	▲
5.	Tubing - check all tubing and connections for signs of water marks/staining where leaking or dripping has occurred.	▲

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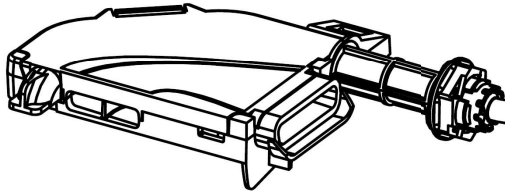
5.4 Hot Water System

6.	Pressure Relief valve - check for leaks. Replace if evidence found.	
7.	Flow meter - check for leaks. NB: If replacing, the unique 4-digit factory calibration value must be configured into the machine through the 'Engineering Mode' menus (see Section 5.6.6)	
8.	Manifold pressure sensor PCB - Check for any signs of leaks in pipework. Refer to previous error log check.	
9.	Pressure switch - Check for scale buildup and leaks.	
10.	Cold tank - Inspect inside cold tank, lid can be removed to check for scale build-up, clean as required and ensure float switch moves freely.	
11.	Manifold assembly - Incorporating 3x solenoids - Jet, injector and return valve. Check for leaks and functionality by using a combination of the air pump, the pressure sensor value and each of the 3 valves (all of which can be operated via Engineering Menu see Section 7.2.2).	
12.	Air pump - Check bin for wet packs. If messy or wet, the de-water cycle has failed. Check air pump motor and one-way valve using the pressure sensor value shown in the Engineering Menu (see Section 7.2.2).	
13.	Cup detect sensors – Clean. If necessary, replace and/or calibrate.	

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5.5 Functional Subassemblies

5.5.1. Pack Heater Bar



The pack heater bar is designed to last the life of the machine.

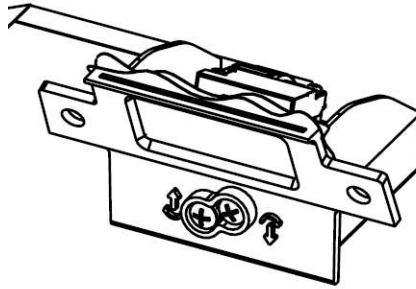
Preventative maintenance

	Action	Critical Check
1.	Jet Nipple - Check for scale, dirt build-up and blockages causing short measures. Disconnecting the Narrow bore pipe from the silicon and unscrewing the heater bar nut (both accessed through the RHS panel), allows removal of the heater bar that can then be rinsed clean.	

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5.5 Functional Subassemblies

5.5.2. Barcode Reader



The barcode reader is designed to last the life of the machine.

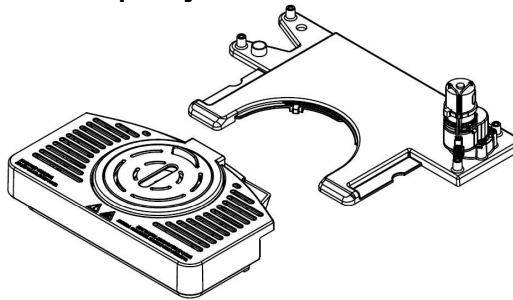
Preventative maintenance

	Action	Critical Check
1.	Barcode Reader – Check barcode reader correctly identifies pack. Check lens for dirt build up. Clean as necessary.	■
2.	Barcode Alignment – Check barcode scanner is aligned to center of the pack during Functional Test.	■

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5.5 Functional Subassemblies

5.5.3. Cup-stand & Drip Tray



The cup-stand and drip tray designed to last the life of the machine.

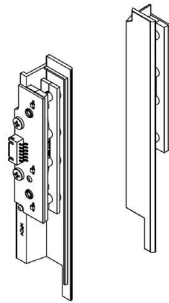
Preventative maintenance

	Action	Critical Check
1.	Drip Tray Rings – Drip tray rings should move cleanly and without resistance or hesitation. Inspect for dirt build up, damage or improper assembly.	☒
2.	Cup-stand motor - Refer to previous error log checks for time outs. NB: If replacing either the assembly or the motor, the type of encoder must be confirmed in the 'Engineering Menus'	☒
3.	Cup-stand drive pin – Pin should engage with drip-tray easily. Inspect for free-motion and sprung return.	☒

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5.5 Functional Subassemblies

5.5.4. Cup Detect



The cup detect system is designed to last the life of the machine.

Preventative maintenance

	Action	Critical Check
1.	Lenses – Inspect for dirt or residue build up and clean if necessary. Using 'Engineering Menus' confirm all three cup detect locations correctly identify obstruction. If the lenses required cleaning, reset the calibration as described in Section 5.6.6.	

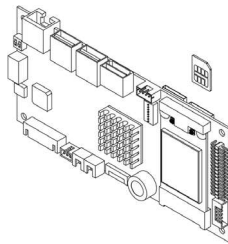
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5.6 Creation 600 Control Boards

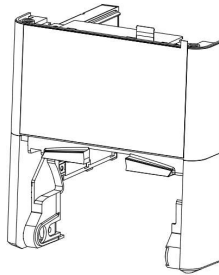
5.6.1 Scope and Purpose

This document covers instructions and what is to be expected when changing any of the FLAVIA® Creation 600 control boards or control board assemblies. Specifically, this covers;

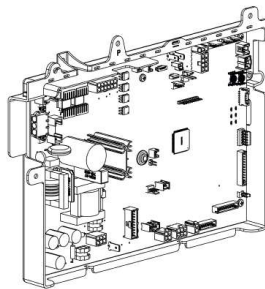
76-24-0781 UI Host Controller (SBC Assembly) including Modem and SIM



76-24-0464 Head Faceplate Assembly including screen and UI Host SBC Assembly



76-24-375 Brewer Controller PCB



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5.6 Creation 600 Control Boards

5.6.2 Glossary of Terms

“Brewer” A FLAVIA® C600 brewer

“SBC” refers to the **Single Board Computer** assembly 76-24-0781 that is also sometimes referred to as the User Interface Host or UI Host. This control board is the main controller that drives the screen, the user interface and all external network connections. The SBC assembly includes a modem and SIM.

“Brewer Controller” refers to the Brewer Controller PCB **76-24-0375** that is the internal brewer mechanism controller.

“IoT” Internet of Things. Specifically, the Lavazza Professional Flavia IoT platform. This is a cloud-based internet platform for data exchange between the brewer and other applications linked to it such as the FLAVIA® IQ IoT portal.

“Serial Number” the serial number of the brewer

“IoT Certificate” A security certificate allocated to the SBC to allow the brewer to access our IoT platform

“Device ID” the unique IoT device ID of an SBC

“Registration” the act of linking an SBC’s IoT Device ID to a brewer’s Serial Number

5.6.3 Introduction to IoT Registration

The brewer is designed by default to connect to the Internet via the built in 3G/4G/LTE Modem and SIM. It is also possible to connect via Wifi or ethernet. Further details are in Section 2.4.

Once connected to the Internet the brewer will try and connect to the Flavia IoT platform. In order to connect to IoT the Brewer must have previously been registered with IoT.

Registering a brewer with IoT is the act of linking the brewer’s Serial Number with the SBC’s Device ID.

For new brewers this Registration is performed on the assembly line during production. If replacing an SBC as part of a repair, then the brewer needs to be re-registered with the Device ID of the new SBC

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5.6 Creation 600 Control Boards

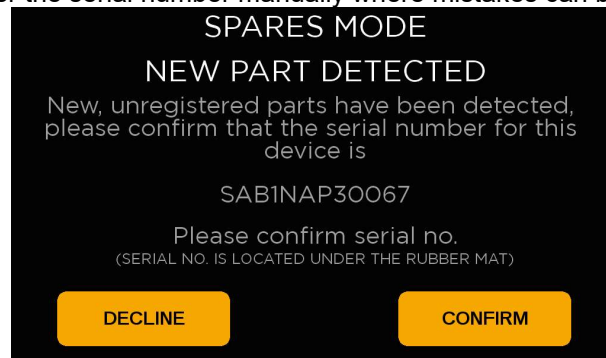
5.6.4 Replacing SBC (76-24-0781) or Head Faceplate Assembly (76-24-0464)

IMPORTANT! When replacing an SBC or head faceplate assembly you cannot re-use an existing SBC that has previously been used. The new SBC must not have been previously registered with another brewer serial number. You must use a new spares assembly that has not previously been registered.

Whenever the brewer is powered on it checks that SBC has a valid serial number and that the serial number in the SBC matches a backup copy stored on the Brewer Controller. If either of these tests fail then the brewer will enter a Spares Mode to check serial numbers and registration status.

5.6.4.1 SPARES MODE – New Part Detected

If a new SBC is fitted that has not previously been used, it will not have a valid serial number. In this case the user will be prompted to confirm the serial number of the brewer. The user will be shown the serial number from the backup stored on the Brewer Controller. This removes the need for the user to enter the serial number manually where mistakes can be made.

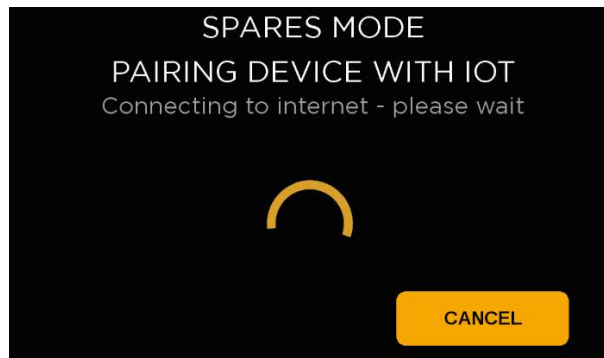


The user should validate the serial number of the brewer (serial number label is under the rubber mat on top of the brewer) and press CONFIRM. Once confirmed the brewer will attempt to connect to the internet and Register the new SBC Device ID and brewer serial number. You will see

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the following screens.



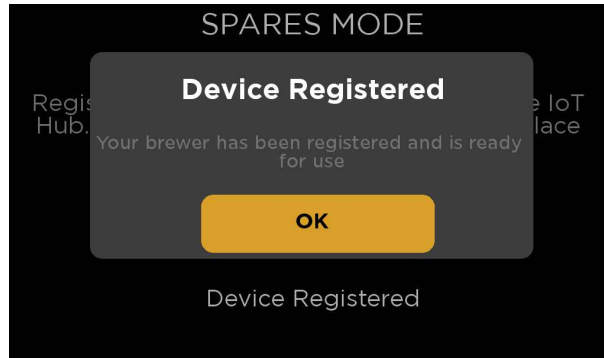
This screen will show whilst the brewer is attempting to connect to the internet if it is not already connected.



When connected to the internet the brewer will register the new SBCs Device ID with the brewer serial number.

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When successfully registered a confirmation is shown. When OK is pressed the brewer will start.

Software updates should be checked for in the Settings->1441->Technician Settings->Flash/Clone or by going through installation mode (Settings->1441->Technician Settings->Operating Modes->Installation) and letting the brewer check for updates.

If the brewer is registered this will show in the Settings->3333->Operator Settings->Networks->IoT Configuration, stating "Registered Yes"

IoT Configuration	
IoT Enabled	☒
Device ID	4bde0faa-22ee-44ef-b04e-7bfd97d5bb67
Registered	Yes
IoT Environment	TEST

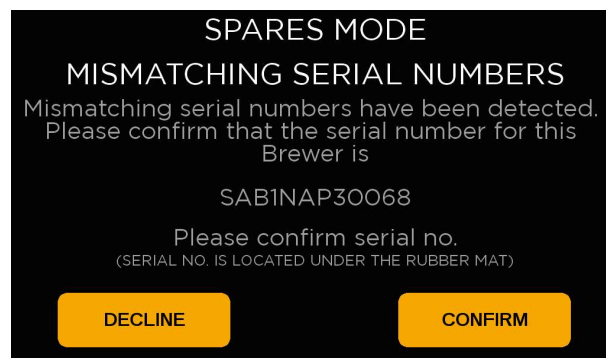
IMPORTANT! If the SBC is replaced then the brewer should be put back into "Installation Mode" for use on site. The EULA must be accepted for usage data and content downloads to function. The installation location details will need to be re-entered.

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5.6.4.2 SPARES MODE – Mismatching Serial Numbers

If, when the brewer turns on, it detects that the serial numbers stored in the SBC and the backup in the Brewer Controller do not match, then the following screen will be shown.



This could happen if the Brewer Controller has been swapped for a new Brewer Controller or a used Brewer Controller that has an old backup of another machine's Serial Number.

In this case the user will be prompted to CONFIRM the Serial Number that is stored in the SBC. If this is the correct Serial Number for the brewer then this will be written to the backup on the newly installed Brewer Controller and the brewer will restart normally.

Note: Brewer Controllers can be re-used as they only contain a backup of a brewer serial number. SBCs cannot be re-used since the SBC's Device ID and brewer serial number have been previously registered.

If, on the other hand, the serial number shown does not match the brewer, then this is an attempt to re-use an existing previously registered SBC.

IMPORTANT!: If the user selects CONFIRM when the serial number shown does not match the brewer then this brewer will have the identity of another brewer on IoT and the data will be incorrect.

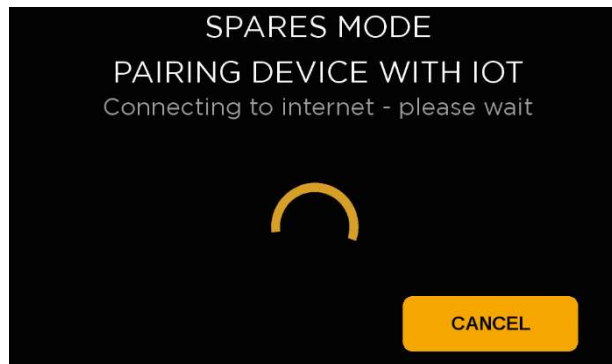
In the case when the serial number displayed does not match the serial number of the brewer user should press DECLINE and replace the SBC with the correct SBC that matches the brewer serial number or a new unused spares SBC.

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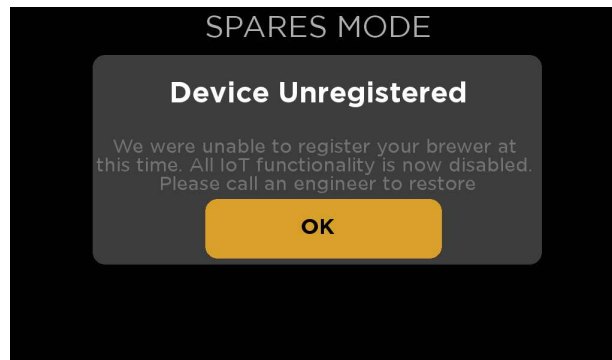
5.6 Creation 600 Control Boards

5.6.4.3 SPARES MODE– Registration Problems

If during the registration attempt the brewer cannot connect to the internet or the registration attempt fails, the user will be shown the screens below.



This screen will be shown indefinitely when the brewer is retrying to connect to the internet. If the brewer cannot connect the user can press CANCEL to abort the internet connection and registration attempt. If the internet connection attempt is cancelled or the registration process fails then the brewer will show the “Device Unregistered” warning.



All brewer IoT functions, including software updates and Touchless phone app operations will be disabled until the brewer is successfully registered.

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If the brewer is unregistered this will show in the Settings->3333->Operator Settings->Networks->IoT Configuration, stating “Registered No”

IoT Configuration	
IoT Enabled	☒
Device ID	4bde0faa-22ee-44ef-b04e-7bfd97d5bb67
Registered	No
IoT Environment	TEST

To avoid the brewer going back through the process of Serial Number Validation and failing again every time the brewer is turned on, this checking will be disabled.

When the brewer is able to successfully connect to the internet once again, then the Serial Number Validation checking can be turned back on and the registration re-attempted. To turn the Serial Number Validation process back on go to Settings->1441->Technician Settings->Operating Modes and tick the “Serial Number Validation” check box. When the brewer is re-started the registration process will check the serial numbers and retry the registration if needed.

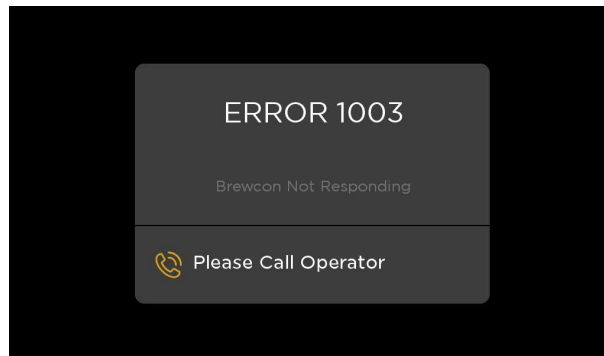
Operating Modes	
Machine Mode	Normal
Pay Mode	Free
Serial No. Validation	☐

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5.6 Creation 600 Control Boards

5.6.5 Replacing the Brewer Controller (76-24-0375)

New Brewer Controller PCBs are supplied without any firmware and will need to be programmed. When fitted and powered up the brewer will show the error:



The Brewer Controller must first be programmed using the menu Settings->1441->Technician Settings->Flash/Clone->Available Firmware x.xx. If the replacement Brewer Controller is not a new blank Brewer Controller, then when the SBC powers up and checks the Firmware Version, it will automatically update the Brewer Controller firmware if needed.

5.6.5.1 Brewer Serial Number Checking

When the Brewer restarts it will note that the Brewer Controller does not have a valid backup of the Serial Number stored in the SBC. The user will see the SPARES MODE MISMATCHING SERIAL NUMBERS warning and confirmation screen (see Section 4.2)

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5.6.6 Important Configuration Checks

Whenever a Brewer Controller is replaced, along with the obvious functional checks that will be needed, there are important settings and calibration operations that should be checked and performed. This includes

1. Flow Meter
2. Cup Sensors
3. Handling Mechanism (Inject and Eject)
4. Water Pump Calibration

Flow Meter Settings

Every flowmeter has been factory calibrated and will have a label stating the 4 digit calibration value.

Enter the Technician settings menus (1, 4, 4, 1) and select "Engineering menus"

Navigate the menu using the "Next" and "Back" buttons to the "Flow Meter uL/Pulse" screen.

Check the setting matches the value printed on the flow meter label. If it does not, then

press the "Edit +" and "Edit -" buttons to adjust the value stated to match that stated on the flowmeter label.

Cup Sensors Calibration

The cup sensors will need re-calibrating with the new board.

Enter the Technician settings menus (1, 4, 4, 1) and select "Engineering menus"

Navigate the menu using the "Next" and "Back" buttons to the "Cup Sense Calibration" screen

Press "Start" and the brewer will calibrate the new emitter receiver pair.

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Handling Mechanism Calibration

It is necessary to calibrate both the pack inject and pack eject speeds.

Enter the Technician settings menus (1, 4, 4, 1) and select "Engineering menus"

Navigate the menu using the "Next" and "Back" buttons to the "H Mech IJ Calibration" screen

Press "Start" and the brewer will calibrate the new handling mech motor Inject speed.

(It is normal to hear the handling mech operate during the calibration process

Navigate the menu using the "Next" and "Back" buttons to the "H Mech EJ Calibration" screen

Press "Start" and the brewer will calibrate the new handling mech motor Eject speed.

Water Pump Calibration

The Water Pump calibration will need resetting.

Enter the Technician settings menus (1, 4, 4, 1) and select "Engineering menus"

Navigate the menu using the "Next" and "Back" buttons to the "Water Pump Calib: Prepare for New pump?" screen

Press the "Start" button so that the brewer can restart the new Water Pump's calibration.

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5.7 Functional Test

The control system requires little or no maintenance beyond a visual check.

Check that the connectors on both control boards are fully seated and that all visible looms are correctly routed and secured in position.

Preventative maintenance

	Action	Critical Check
1.	Pack bin/chute – Check machine's pack waste chute is aligned to worktop hole. Check bin/chute switch operation.	
2.	Calibrations – Where parts are either out of spec or have been replaced: • Handling Mechanism – Inject & Eject time calibration • Bar code reader – Central to pack bar code • Pump – Register replacement in settings and auto-calibrate • Cup sensor – Run an auto calibration • Flow meter – Unique 4-digit factory calibration value must be configured into the machine!	
3.	Vend-Look-Listen: • Cup stand motor – remove cup stand and replace. Motor will re-calibrate. Check for smooth operation. • Cup detect sensors. • Bar code reader - test using Bar-Coded pack. • Touch screen operation. • Pack opening • Cup volumes using nozzle - Hot Chocolate target 163.5g (min 158.8 – max 168.2) • Drink temperature 76 - 84°C in a paper cup.	
4.	Complete history card (UK only) - recording all addition work, Portable Appliance Test results and filter replacements.	
5.	KEY OPERATOR TRAINING: Ensure that importance of hygiene and regular cleaning is fully understood and appreciated by the Operator	

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Section 6

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- 6.2** Pack Handling Mechanism
- 6.3** Door, Door Latch and Pack Barrier
- 6.4** Drip Tray and Drive
- 6.5** Water System
- 6.6** PCBA and Head-faceplate

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6.1 Overview

Introduction

This section contains a quick reference guide to potential maintenance procedures not otherwise covered in this technical manual.

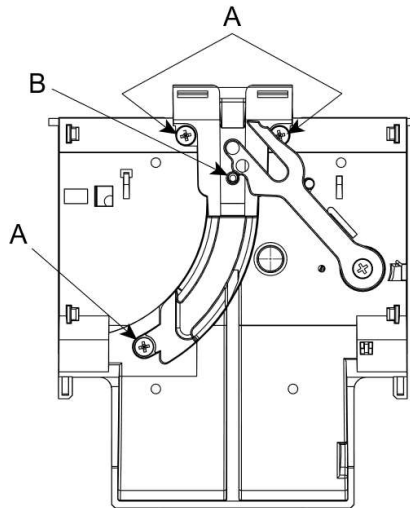
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6.2 Pack Handling Mechanism

6.2.1. Pack Rail

The pack rail requires infrequent cleaning to remove small deposits which can build up over time. The rate of build-up is increased when used in conjunction with a Flavia Chill Module.

To effectively clean the pack rail, the rail should be taken out of the brewer by removing the three securing screws marked 'A' below. The pack injector marked 'B' should be pressed upward using a small screwdriver to release the rail front. The rail front can then be slid forwards and out of the system. Once the rail front has been removed, the rail rear rail can be lowered vertically away from the mechanism.



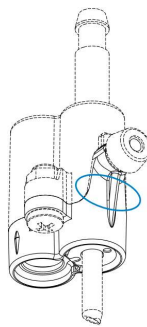
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6.2 Pack Handling Mechanism

6.2.2. Injector Carriage

PCO388 – Pack Eject Timeout

An increased number of pack eject timeouts can indicate a failure of the injector carriage. This would normally only be seen if the C600 is installed with Flavia Chill. The machine will continue to work for a period with a failed carriage. The most likely place for a failure is on the feature indicated below:



6.2.3. Handling Mechanism Calibration

If the Handling Mechanism is repaired or replaced it will require recalibration using the engineering menu (See Section 7.2). The process is as follows

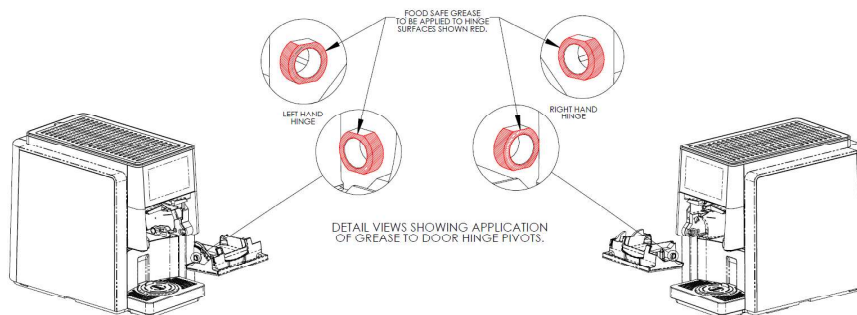
- Ensure the Brewer is powered on and that there are no packs on the pack rail.
- Enter the Technician Menu (1-4-4-1)
- Select Engineering Menu
- Cycle through the menu to [H-Mech IJ Calibration](#)
- Press 'Start' – the Handling Mechanism will run for a short period of time and stop automatically once the calibration is complete.
- Cycle through the menu to [H-Mech EJ Calibration](#)
- Press 'Start' – the Handling Mechanism will run for a short period of time and stop automatically once the calibration is complete.
- Exit the menus – Calibration is now complete

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6.3 Door and Door Latches

6.3.1. Door Pivot

In cases where the door is slow to open or fails to fully open for a vend, food safe grease (72-31-0366) can be applied to the door in the positions shown below.



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6.4 Cup Stand and Drip Tray

6.4.1. Drip Tray Cleaning

The drip tray must not be cleaned in a dishwasher as this can lead to damage of both the cosmetic finish and the functional elements of the automatic cup stand. Distortion of the cup-stand core will prevent the cup stand moving smoothly and, in extreme instances, could cause a '[CUP STAND FAULT](#)' (PCO 341).

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6.5 Water System

6.5.1. In-line Scale Filter

The C600 brewer **MUST** be fitted with an in-line scale and taint filter as outlined in the user manual. The system must not be descaled and the procedure used for previous products (e.g. Flavia C500, Flavia C400, etc.) is not compatible with this brewer.

The following symptoms are indicative of a filter not being used or replaced appropriately:

- Dripping from the Injector position
- Low drink volumes
- Incorrect pressure sensor reading
- Failure of the air pump leading to internal leaks and failure to dewater packs
- 'Water Cooling' message due to boiler over-heating

6.5.2. Boil Water Notice

If a 'Boil Water Advisory' notice is issued by the mains water supplier, the internal particulate filter should be replaced and the machine flushed by following the procedure below:

- Enter the engineering menu (1-4-4-1)
- Disable the hot tanks
- Power off the brewer
- Remove the drip tray and pack bin.
- On the inside right of the pack bin area, locate the red cold tank drain. Unclip the red cold tank drain cap, unplug the drain tube, and drain the contents of the cold tank into the empty pack bin. Dispose of the pack bin water.
- Replace the cold tank drain cap and tube
- Re-fit the pack bin and cup stand.
- Remove the rubber mat on top of the brewer. Locate the red filter cover cap, loosen the filter cap screw, and turn the cap counter clockwise to remove.
- Remove the internal filter and dispose of the old filter. Insert the new filter (Part# 76-24-0634), place the cap in the unlock the position, press down, turn clockwise to the lock position, and re-tighten the filter cap screw.
- Turn the C600 on via the power switch.

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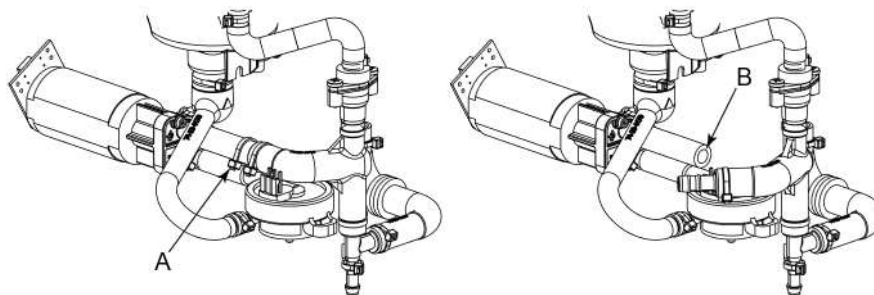
6.5 Water System

- The brewer will prompt you to “Prime and Flush”. Follow the onscreen prompt pulling the pack bin out in line with the pack door, and press continue (please note: the brewer will now prime and flush the water system causing water to flow into the pack bin).
- Once the flush has completed, empty the water from the pack bin and replace the pack bin and drip tray. The hot tanks will heat and the screen will return to the drinks carousel to begin brewing.

6.5.3. Installation PCO377

During installation, the brewer may display the error PCO377 – Flow Meter Timeout. This can be fixed in the majority of cases by following the procedure below:

- Switch off and isolate the machine
- Remove the right hand cosmetic and safety side panel
- Removing the cable tie at the point marked 'A' in the figure below and removing the straight hose connector as shown in the right hand image
- Remove and re-insert the particulate filter in the cold water tank until water is visible in the hose marked 'B'
- Reinsert the connector in its original position and secure with a cable tie
- Replace the side covers and re-power the brewer
- The brewer will now prime normally

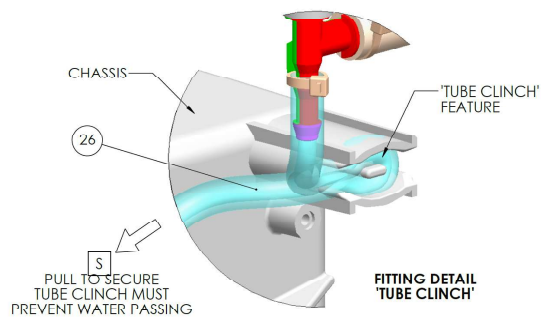


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6.5 Water System

6.5.4. Drain Hose

The Hot-Tank drain hose is accessed through the right hand safety panel and is secured in a chassis feature when not in use to prevent leaks and reduce drink volume variability. To re-secure the hose after use it should be fed through the switch-back feature as shown below and then the free end should be gently pulled tight to form a seal.

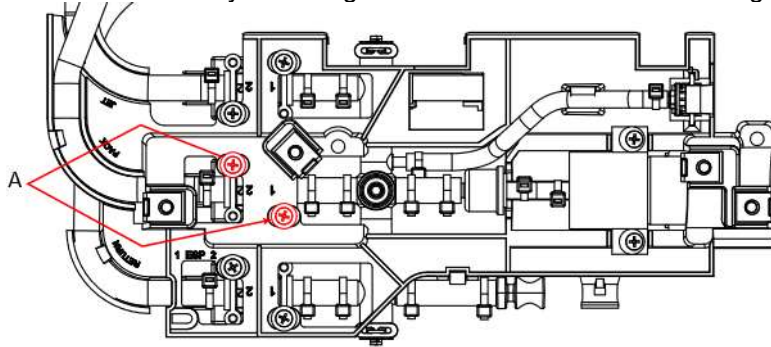


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6.5 Water System

6.5.5. Injector Valve

If the brewer is used with a Flavia CHILL module, the injector valve should be replaced after 30,000 vends as preventative maintenance. The injector valve is released by removing the screws marked 'A' in the image below:



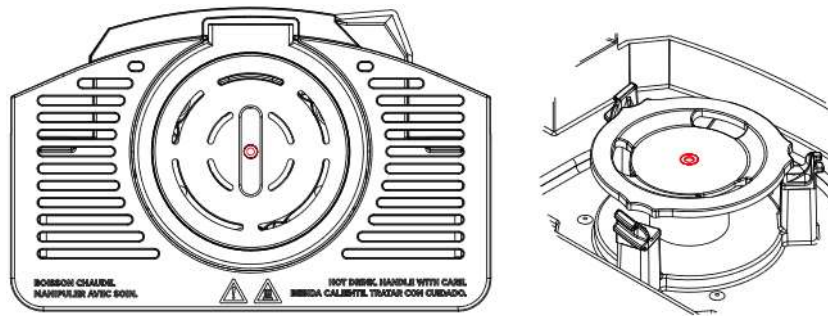
When fitting a new valve, make sure that the embossed '1' and '2' on the valve are in the orientation marked on the valve bracket. If needed, the solenoid body can be rotated to align the electrical connections correctly.

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6.5 Water System

6.5.6. Heater Bar and Jet replacement

If the Heater Bar requires replacement, care should be taken to avoid overtightening the nut that secures the assembly to the chassis. Once the new bar is fitted, jet alignment should be checked by placing a transparent cup on the cup-stand and manually activating the jet using the engineering menu (1-4-4-1). Correct Jet alignment will cause water to strike the cup above the target moulded into the cup stand drive and highlighted red in the picture below.



6.5.7. Internal Particulate Filter Access Cap

The particulate water filter access cap on the top of the brewer is secured using a bayonet fitting. Accidental rotation is prevented with a M4 screw located in the lid. To remove the cap this screw must be partially removed, allowing the lid to rotate.

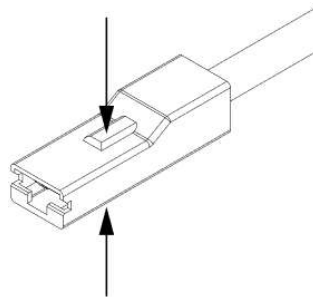
To re-secure the cap, the screw need only be turned until its head is flush with the cap, it does not need to be tightened to function. Overtightening this screw can produce white stress-marks in the plastic cap but this will not affect performance.

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6.6 Electrical System

6.6.1. Boiler connections

To remove the boiler electrical power connections, the black plastic housing must be squeezed in the location shown below and slid back to release the securing latch. This is not necessary for re-assembly.



6.6.2. Main PCBA connector housings

Removal of SBC Comms loom from the connector on the Main Control Board can result in the female connector housing being pulled away from the board surface, essentially moving up the pins soldered to the board. This results in the pins effectively being shortened within the connector housing and thus not making a poor or no connection to the male connector on the SBC comms loom when fitted. This manifests as the brewer hanging at the boot up screen and is resolved by pushing the connector housing back down onto the surface of the board.

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6.6 Electrical System

6.6.3. Barcode reader

If the Barcode Reader assembly has been replaced, or the brewer is failing to read barcodes, you can check the alignment of the laser against reference marks on the inside of the brewer chassis. The procedure to do this is as follows:

- Remove the door from the brewer
- Enter the Engineering Menu
- Navigate to the 'Barcode Alignment' screen and press the 'Test' button. This will activate a wide angle beam which will reach the side of the brewer.
- Compare the position of the beam with the top of the alignment marks in the brewer
- If the laser is too high, loosen the right hand screw and tighten the left hand screw the same amount
- To move the laser down, reverse this operation.
- Press 'Exit Menus' when done

6.6.4. Connecting to IoT in areas of poor cellular reception

By default, the brewer will use the built-in modem for cellular connectivity to the internet. If there is poor, or no AT&T cellular signal, then a suitable available internet WiFi connection can be selected. The WiFi network selected must allow direct internet connection with the use of a passkey and not via an online portal web page that requires a login. Check with a local IT admin if in doubt.

The Network Status Screen shows the current network adapter being used (Cellular, WiFi or LAN) and the status of the Internet and IoT connections.

To see the status screen, go to Settings->3333->Networks

If there is a RED cross above the Internet Connection press "Modem Settings" and check the signal strength.

If there is poor or no signal then you may need to try an external antenna or switch to WiFi or LAN connection. See instructions below or reference Doc-000715.

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6.6 Electrical System

Swapping to connect via WiFi

To connect the brewer to the internet using a WiFi connection, follow the steps below:

- Disable the modem by going to Settings->3333->Networks->Modem Settings and uncheck “Enable Cellular Modem”
- Go to Settings->3333->Networks->WiFi Configuration and check “WiFi Enabled”.
- When the WiFi is enabled press “Show Networks” and select the WiFi network from the list of available networks
- Enter the PassKey for the Network and check the “Connect to Network” check box

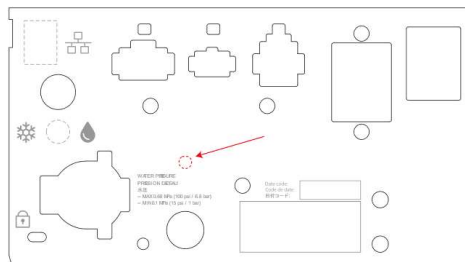
The brewer will now attempt to connect to the selected WiFi network.

Fitting an optional external antenna

Note: this requires the optional accessory – 76-10-5575 – Aerial externalization kit.

To fit this accessory follow the steps below:

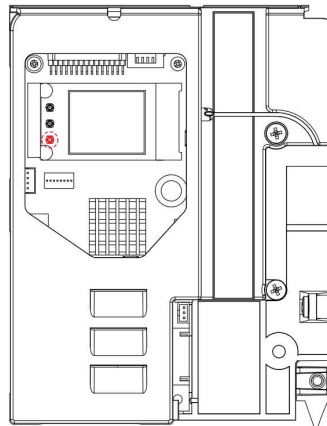
- Power off and isolate
- To gain access to the necessary areas, Remove the left-hand side cosmetic side cover, left hand side access panel, filter access screw cap, top cover and rear cover.
- On the rear cover label located on the rear, at the bottom of the brewer, carefully cut through the label in the area indicated by a dashed circle



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- From inside the C600, Insert the bulkhead connector on the aerial externalisation loom through the hole exposed in the rear panel and secure in place with the 2 washers and nut supplied in the Kit
- On the SBC (accessed through the Left-hand side cover), remove the existing antenna connection on the lower of the 3 modem connectors and connect the externalisation loom in its place.



- Replace all covers removed and locate the aerial in a location with good cellular network reception.
- Replace all covers removed and locate the aerial in a location with good cellular network reception.

Your C600 should now have a connection to the internet which can be confirmed as described above.

Connecting via Ethernet cable

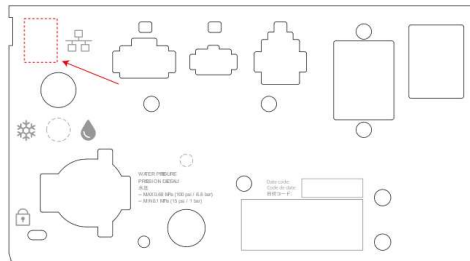
Note: this requires the optional accessory – 76-10-5567 – RJ45 Upgrade kit

If there is no available cellular or WiFi wireless signal you can connect to the internet via Ethernet. The Ethernet connection must connect to a router with internet access. You may need to check with a local IT admin to make sure that the connection being used has an open connection to the internet that does not require a login or password.

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6.6 Electrical System

- Disable the modem by going to Settings->3333->Networks->Modem Settings and uncheck "Enable Cellular Modem"
- Disable WiFi by going to Settings->3333->Networks->WiFi Configuration and uncheck "WiFi Enabled".
- Power off and isolate
- To gain access to the necessary areas, Remove the left-hand side cosmetic side cover, left hand side access panel, filter access screw cap, top cover and rear covers.
- On the rear cover label located on the rear, at the bottom of the brewer, carefully cut through the label in the area indicated by a dashed rectangle.



- Clip the RJ45 panel connector into the aperture that you have created
- Inside the brewer, use the Ethernet cable to connect from the SBC (accessed through the left-hand side panel) to the bulkhead connector that you have just fitted.
- Replace all covers removed.
- Connect the new RJ45 connector on the rear of the brewer to an ethernet cable with an active internet connection (not provided in kit)

Your C600 should now have an active connection to the internet. This can be confirmed as described above.

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Reference Information

Contents

- 7.1** Operator Mode
- 7.2** Engineering Mode
- 7.3** Error Messages and Events

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7.1 Operator Menu

This section describes the menus available in 'Operator Mode'. These menus are one of three sets of menus available in the 'Menu Mode' of operation of the FLAVIA® Creation 600.

The Brewer has two modes of operation: '**Vend Mode**' (for drinks vending) and '**Menu Mode**' (for operator/engineer audit, setup and test). The Brewer always powers-up into Vend Mode; **to enter Menu Mode**, press the top left corner of the screen for at least 1 second and the LCD display will then change to show the default audit menu. To exit Menu Mode, press the Arrow in the top left of the screen.

To access the 'Operator Settings' menu, enter the 'Operator Mode' password. The default 'Operator Mode' password is '3333' and press 'OK'. This default password can be changed to a user-defined 4-6 digit number - see Section 7.1.4. If the wrong password is entered, press the 'Del' key and retry (there is no need to exit the menu screen first).

The Operator Settings menu allows the key operator to read the Brewer's audits, to reset its interim audits and to perform maintenance tasks.

The menus are laid out in a contextual graphical interface and can be navigated by touching the relevant icon. To return to the previous menu, press the left facing arrow in the top left corner of the screen.

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7.1 Operator Menu

7.1.1 'Operator Settings' Menu Tree

- Regions & Languages
 - Country
 - Consumer Language
 - Alt. Consumer Languages
 - Operator Language
 - Time Zone
- Time & Date
 - Show Clock
 - Date
 - Time
- Site Preferences
 - Cups
 - Default Volume
 - Max Hot Volume
 - Cup Stand
 - Mode
 - Pack Bin
 - Pack Bin
 - Max Packs in Bin
 - Modes
 - Touchless Mode
 - Demo Mode
 - Days until Demo Mode ends
 - Days to Go Count
 - Vends until Demo Mode ends
 - Number of vends to Go
 - Reset
 - System Settings
 - Screen Brightness
 - Operator Password
 - Schedules
 - Not Yet Implemented
 - Drinks & Packs
 - Milk Drink Options
 - Drinks To Show
 - Pack Types Available
 - Show Clean Barcode Alert
 - Energy Saving
 - Energy Saving Timeout

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7.1 Operator Menu

- IoT Settings
 - Report Low Stock
 - Report Faults
 - Report Mess
 - Show IoT Content
 - IoT Content Check
 - Content Mgt...
 - Content List
 - Delete
 - IoT Software Check
- Maintenance
 - Flush Brewer
 - Clean Cup Sensors
 - Clean Barcode Sensors
 - Assemble Drip Tray
- Networks
 - Network Status
 - Modem Settings
 - Enable Cellular Modem
 - APN Endpoint
 - Modem ID
 - SIM ICCID
 - Signal Strength
 - Data Rate (B/s)
 - IoT Configuration
 - IoT Enabled
 - Device ID
 - Registered
 - IoT Environment
 - WiFi Settings
 - WiFi Enabled
 - Network-Interface State
 - Signal Strength
 - Current Network
 - SSID
 - PassKey
 - Connect to Network
 - Show Networks

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7.1 Operator Menu

Location

- Building No./Name
- Floor No.
- Location Type
- Description
- PCO Contact

Sales & Audits

Interim Audits

- Number of Vends
- Number of Packs
- Reset

Running Audits

- Number of Vends
- Number of Packs

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7.1 Operator Menu

7.1.2 Regions & Languages

Country	Set the country in which the brewer is located
Consumer Language	Set the language to be displayed as default in 'Vend Mode'
Alt. Consumer Languages	Set the alternate languages to be available in 'Vend Mode'
Operator Language	Set the language to be displayed in 'Operator Mode'
Time Zone	Set the local time-zone

7.1.3 Time & Date

Show Clock	If checked, a digital clock will display in 'Vend Mode'
Date	Set the current date in the format YYYY/MM/DD
Time	Set the current time

7.1.4 Site Preferences

Cups

Default Volume	Set the default drink volume (fl. Oz) for a Hot vend
Max Hot Volume	Set the maximum drink volume (fl. Oz) for a Hot vend

Cup Stand

Mode	Auto: Enable the automatic cup stand Off: Disable the automatic cup stand
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Pack Bin

Pack Bin	Internal: Use the supplied pack bin to store a set number of used Freshpacks inside the brewer External: For use with a pack bin funnel and an external pack bin Internal (Smart Bin Count): Use the supplied pack bin to store used Freshpacks inside the brewer. Bin capacity is adjusted based on the preceding vend types.
Max Packs in Bin	This menu will only appear if 'Internal' is selected in the 'Pack Bin' menu. Define the maximum number of packs in the bin before requiring the bin to be emptied.

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7.1 Operator Menu

Modes

Touchless Mode	If checked, the use of the TapN'Brew mobile app is enabled. The brew session QR code will be displayed in the bottom right of the 'Vend Mode' carousel.
Demo Mode	When in demo mode the brewer will display different "while brewing" content that is geared towards new users. Number of Days : Enable 'Demo Mode' for a period defined in days. Number of Vends : Enable 'Demo Mode' for a period defined in vends. Days or Vends : Enable 'Demo Mode' for a period defined as the earliest of either days or vends. On : 'Demo Mode' is enabled indefinitely. Off : 'Demo Mode' is disabled.
Days until Demo Mode ends	This menu will only appear if either ' Number of Days ' or ' Days or Vends ' is selected. Define the number of days to enable 'Demo Mode'.
Days to Go Count	This menu will only appear if either ' Number of Days ' or ' Days or Vends ' is selected. Display how many days remain on the 'Demo Mode' counter
Vends until Demo Mode ends	This menu will only appear if either ' Number of Vends ' or ' Days or Vends ' is selected. Define the number of vends to enable 'Demo Mode'.
Number of vends to Go	This menu will only appear if either ' Number of Vends ' or ' Days or Vends ' is selected. Display how many days remain on the 'Demo Mode' counter
Reset	Confirm changes and reset the duration counters

System Settings

Screen Brightness	Allows the adjustment of screen brightness. Acceptable values are between 2 (lowest) and 10 (brightest).
Operator Password	Enter the desired Operator Password . This must be between 4 and 6 digits long.

Schedules

Not Yet Implemented

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7.1 Operator Menu

Drinks & Packs

Milk Froth	Select which sources may be used for Frothed Milk drinks. Milk Froth Packs: Only milk supplied in FreshPacks may be used Liquid Milk: Only milk supplied in a consumer's cup may be used Milk Froth Packs and Liquid Milk: Both sources of Milk may be used Lavazza 1-Pack or Liquid Milk: Lavazza 1-Pack Specialities or milk supplied in a consumer's cup may be used. Lavazza 1-Pack: Only 1-Pack specialities may be used None: If no form of milk pack, or milk-based drinks are available on site. All milk-based drinks will be removed from the carousel.
Drinks	Select the drinks which will be visible in the 'Vend Mode' carousel
Pack Types Available	Select which FreshPacks are available on the site
Show Clean Barcode Alert	Display an alert after a number of unsuccessful FreshPack barcode read attempts.

Energy Saving

Energy Saving Timeout	Set the duration of inactivity after which the brewer will enter Energy Saving mode. In normal operation the brewer will maintain a ready-to-vend water temperature by periodically powering the boilers. In energy saving mode, the water temperature will be allowed to fall. On waking, the brewer will display a please wait screen while the boilers come up to temperature.
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7.1 Operator Menu

IoT Settings

Report Low Stock	Enable low stock reports to be sent via IoT
Report Faults	Enable brewer fault reports to be sent via IoT
Report Mess	Enable brewer mess reports to be sent via IoT
Show IoT Content	Enable the display of content delivered via IoT
IoT Content Check	Enable periodic checking for new content via IoT
Content Mgt	Displays a list of content stored on the brewer. To delete specific items, check the desired content and select 'Delete'
IoT Software Check	Enable periodic checking for new software versions via IoT

7.1.5 Maintenance

Flush Brewer	Begin a brewer flushing cycle. Once pressed, a confirmation menu will be shown. Following confirmation, water will be flushed through either the jet or the injector. Ensure a receptacle of >2l is in position to collect the flushed water. It is recommended to run this process twice to flush both water paths. The brewer's empty pack bin can be used as a convenient receptacle for this by moving it forwards to the vend area.
Clean Cup Sensors	Display the cleaning sequence for the cup detect sensors.
Clean Barcode Sensors	Display the cleaning sequence for the barcode reader lens
Assemble Drip Tray	Display the assembly sequence for the drip tray.

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7.1 Operator Menu

7.1.6 Networks

Network Status

Network Type	The status of the network connection is shown with either: a red cross – indicating no connection; or green tick – indicating a successful connection. The type of connection is shown below the brewer graphic: None: No Connection Cellular: The brewer is connected by the internal cellular modem WiFi: The brewer is connected to a local wireless network
Internet Connection	The status of the connection to the Internet is shown with either: a red cross – indicating no connection; or green tick – indicating a successful connection.
IoT Connection	The status of the connection to the IoT service is shown with either: a red cross – indicating no connection; or green tick – indicating a successful connection.

Modem Settings

Enable Cellular Modem	Enable the Cellular modem
APN Endpoint	IoT Access Point
Modem ID	Modem serial number
SIM ICCID	Sim ID number
Signal Strength	Signal strength is indicated by a 4-bar graphic with bars filling from grey to green.
Data Rate (B/s)	Current achievable data rate

IoT Configuration

IoT Enabled	Enable the IoT service
Device ID	Displays the unique Device ID
Registered	Indicates if the brewer is registered with the IoT service
IoT Environment	Displays the IoT service the brewer is connected to. This should read ' PROD '

WiFi Configuration

WiFi Enabled	Enable wireless network connectivity
Network-Interface State	Indicates the current network adaptor state – this can be either ' Up ' or ' Down '

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7.1 Operator Menu

Signal Strength WiFi Info:	WiFi signal strength is indicated by a 4-bar graphic with bars filling from grey to green. Wireless network details are displayed if available.
Current Network	
SSID	Selected wireless network ID
PassKey	Press to enter the Pass Key for the selected wireless network
Connect to Network	Enable to connect to the network with the name and passkey entered
Show Networks	Press to scan for available networks

7.1.7 Location

Building No./Name	Enter the name or number of the street address where the brewer is located.
Floor No.	Enter the floor ID where the brewer is located.
Location Type	Select the type of location the brewer is located in.
Description	Enter a description which can be used to identify the brewers location within the building/floor.
PCO Contact	Enter the contact details of the key contact in the event of a brewer error.

7.1.8 Sales & Audits

Interim Audits

Number of Vends	A counter indicating the number of total vends (excluding hot water) since the counter was last reset.
Number of Packs	A counter indicating the number of total packs used since the counter was last reset.
Reset	Press to reset the Interim Audit counters

Running Audits

Number of Vends	A non-resettable counter indicating the number of total vends (excluding hot water).
Number of Packs	A non-resettable counter indicating the number of total packs used.

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7.2 Technician Settings

This section describes the menus available in 'Engineering Mode'. These menus are one of three sets of menus available in the 'Menu Mode' of operation of the FLAVIA® Creation 600.

The Brewer has two modes of operation: '**Vend Mode**' (for drinks vending) and '**Menu Mode**' (for operator/engineer audit, setup and test). The Brewer always powers-up into Vend Mode; **to enter Menu Mode**, press the top left corner of the screen for at least 1 second and the LCD display will then change to show the default audit menu. To exit Menu Mode, press the Arrow in the top left of the screen.

To access the 'Technician Settings' menus enter Menu Mode (as described above) and enter the 'Engineering Mode' password '1441' and pressing '**OK**'. If the wrong password is entered, press the 'Del' key and retry (there is no need to exit the menu screen first).

7.2.1 'Technician Settings' Menu Tree

Engineering Menus

- Creation 600 Firmware:
- Language (Technician):
- Cup Detection:
- Cup Sense Calibration:
- Cup Tx Osc Period:
- Cup Sense 3 Adjust:
- Cup Sense 2 Adjust:
- Cup Sense 1 Adjust:
- Cup Detectors:
- Espresso Module:
- Pack Bin Type:
- Maximum Packs in Bin:
- Water Supply:
- FLAVIA CHILL Module:
- Bar Code Scan Engine:
- Scan Engine Version:
- Automatic Cup Stand:
- Hot Air Pack Opening:
- Flow Meter uL/Pulse:
- Host Serial Number:
- Vend Mode:
- Inter-Vend Delay /sec:

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7.2 Technician Settings

Inj Water Pump Speed:
Jet Water Pump Speed:
Air Pump Speed:
Water Pump Calib:
Pack Door:
Bar Code Scanner:
Bar Code Alignment:
Pack Door Cycle Time:
Pack Door Motor:
Pack Door Motor:
Handling Mech:
Reverse H-Mech Motor:
Inject Time /ms:
Eject Time /ms:
H-Mech IJ Calibration:
H-Mech IJ Cycle Time:
H-Mech EJ Calibration:
H-Mech EJ Cycle Time:
Pack Bin Detector:
Drip Tray Detector:
Cup Stand Calib State:
Cup Stand Calib Time:
Cupstand Encoder Type:
Cup Stand Encoder:
Cup Sense Inputs /mV:
Cup Detectors:
Cup Detection System
Cup Stand Motor:
Pack Heater:
Priming Valve:
Injector Valve:
Jet Valve:
Water Pump Motor:
Air Pump Motor:
Pressure Sensor /psi:
Pressure Switch:
Flow Meter Status:
Pump Metered Water:
Internal Cold Tank:
External Water Tank:
Chiller Air Pump:
Chiller Water Pump:
Chiller Status:

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7.2 Technician Settings

PSU 5V Supply:
PSU 24V Supply:
Pack Fan Status:
Pack Fan Speed (Hz):
Pack Heater:
Hot Tank Test Mode:
Pre Tank Status:
Main Tank Status:
Hot Tank Control:
Hot Tank Flush:
Hot Tank Drain:

Modules

FLAVIA® CHILL
FLAVIA® REFILL

Resets

Restore Defaults

Operating Modes

Machine Mode
Serial No. Validation

Flash/Clone

Software Up To Date/Software Available X.X.X.X
Firmware Up To Date/Firmware Available X.XX
Clone From Key
Clone To Key
Copy Database
Eject USB Key

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7.2 Technician Settings

The menus can be cycled through by repeatedly pressing the Middle Left ('Next') button; to return to a previous menu press the Top Left ('Back') button.

Each menu consists of a title (ending in a ':') on the first line followed by its 'value' on the second line. To change a value, press the Top Right or Bottom Right buttons as required.

Pressing the 'Next' button on the last 'Engineering Mode' menu will wrap-around to the first menu (and vice-versa for the 'Back' button).

The language in which these menus are displayed is determined by the 'Language (Technician):' configuration menu described below.

7.2.2 Engineering Mode

Creation 600 Firmware:	Displays the installed Firmware version
Language (Technician):	Sets the Language for this menu system
Cup Detection:	Enabled: Enables the cup detection system Disabled: Disables the cup the cup detection system. If cup detection is disabled, then the automatic cup stand will not operate correctly and the brewer will commence a vend on request regardless of the absence of a cup.
Cup Sense Calibration:	Begin the recalibration process for the cup detect array. Ensure there is no obstruction between the cup detect lenses and that the lenses are clean before beginning this process. The value of this menu item is the cup detect value of each of three cup detects in the order: top, middle, bottom. A higher value indicates an obstruction.

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7.2 Technician Settings

Cup Tx Osc Period:	This menu sets the period/frequency of the cup sensor transmitter pulse train, with the aim of matching the period/frequency to which the cup sensor receiving circuit is tuned. This value should not require adjusting directly - the automated calibration process, via the "Cup Sense Calibration:" menu, should be capable of setting a suitable value. Note that a value of 195 (e.g. in a new BrewCon PCBA) signifies that the cup sensors have never been (successfully) calibrated; in this state, the consumer interface will be showing a "clean cup sensors" screen. The 3 figures in parentheses are the average voltages, in millivolts, being read from the 3 cup sensor circuits in the order Top, Middle, Bottom. They show the effect that this menu value is having on the cup detect circuitry, should manual calibration ever be necessary.
Cup Sense 3 Adjust: Cup Sense 2 Adjust: Cup Sense 1 Adjust:	This menu sets a fine-tuning adjustment for the output from the Top cup sensor receiving circuit (to cope with component tolerances and degradation). This value should not require adjusting directly - the automated calibration process, via the "Cup Sense Calibration:" menu, should be capable of setting a suitable value. Note that this value can also be altered by the automatic fine-tuning that is carried out by the software when the brewer becomes idle after a vend*. The 3 figures in parentheses are the average voltages, in millivolts, being read from the 3 cup sensor circuits in the order Top, Middle, Bottom. They show the effect that this menu value is having on the cup detect circuitry, should manual calibration ever be necessary.
Cup Detectors:	This menu item gives a visual indication of the status of each cup detect sensor. A 'Y' indicates a cup is detected, a 'N' indicates no cup detected, and a 'Y?' indicates an illogical cup detection (e.g. middle sensor but not lower sensor).
Pack Bin Type:	An indicator showing which bin type is selected. This is editable in the Operator Settings menu (Section 7.1.3)

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7.2 Technician Settings

Maximum Packs in Bin:	An indicator showing the allocated bin capacity. This is editable in the Operator Settings menu (Section 7.1.3)
Water Supply:	An indicator showing the source of water for the brewer. This is editable in the Technician Settings menu. The options are: Plumbed-in External Water Tank
FLAVIA CHILL Module:	An indicator showing if the CHILL Module functionality is enabled. This is editable in the Technician Settings menu. The options are: 'Not Fitted' 'Fitted (2' Water Pipe)' 'Fitted(7' Water Pipe)'
Bar Code Scan Engine:	This menu item informs the firmware which barcode reader is installed in the brewer.
Scan Engine Version:	An indicator showing the version of barcode reader scan engine installed in the brewer.
Automatic Cup Stand:	An indicator to show if the automatic cup stand is enabled.
Hot Air Pack Opening:	This menu item allows the hot air pack opening system to be disabled if required. The only reason to disable this feature would be when waiting for faulty Hot Air Pack Opening system components to be replaced.
Flow Meter uL/Pulse:	This menu configures the volume of water dispensed in microlitres (µL) per flow meter pulse so that the software can calculate the volume of water pumped. If the flow meter requires replacing, this value should be updated to match the four-digit code found on the label on the side of the flowmeter. This menu item is not appropriate to tailor drink volumes to suit a customer's preference. If the customer requests larger or smaller drink volumes then the appropriate drink volumes should be adjusted in the Site Preferences menu. If this menu is mis-used then it will cause undesirable side-effects with pack-opening, mess and system pressures.
Host Serial Number:	An indicator of the brewer host serial number

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7.2 Technician Settings

Vend Mode:	This menu selects the mode in which the Brewer will vend when Vend Mode is next entered. This setting can be changed to one of: 'Normal', 'Test', 'Automatic' or 'Automatic + Pack Door'. For normal vending operation, the mode should be left at its default setting of 'Normal'. The 'Automatic' and 'Automatic + Pack Door' vend modes are for R&D test purposes only. The 'Test' vend mode allow the Brewer to vend without the the normal requirements on packs, pack door, pack bin, drip tray, cup stand, cups, chiller readiness. This allows troublesome parts to be circumvented so that the rest of the system can be tested. This mode also inhibits updates to the pump speed calibration figures and the PCO/Event Logs so that the testing has no lasting effects. As a safety measure, the vend mode will be reset to 'Normal' at the end of a successful test vend.
Inter-Vend Delay /sec:	This menu will only be displayed if the 'Vend Mode:' is set to 'Automatic' or 'Automatic + Pack Door'. This setting and the use of these modes are for R&D test purposes only and should not be used.
Inj Water Pump Speed:	An indicator to show the current water pump speed used when delivering water through the injector valve. Due to variations in the system, this value is not directly comparable between brewers.
Jet Water Pump Speed:	An indicator to show the current water pump speed used when delivering water through the jet valve. Due to variations in the system, this value is not directly comparable between brewers.
Air Pump Speed:	This menu item controls the speed of the Air Pump. This should be set to 100.

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7.2 Technician Settings

Water Pump Calib:	This menu informs the brewer that a new pump is to be installed and will allow the pump calibration to adjust more rapidly than normal. The pump calibration is distinct for injector and jet dispensing and so both must be carried out. The recommended calibration vending is: 1x Barcoded coffee (7oz) 3x Choco (these can be nozzles only) The menu may read: "Prepare for New Pump ?" "Press START to Confirm" "Calibration Prepared"
Pack Door:	The menu displays the current states of the pack door input ('Open' or 'Closed') and the pack detect input ('No Pack' or 'Pack Present'). With the pack door closed and no pack present, the menu should display 'Closed, No Pack'. If the pack door is opened (either manually or by pressing the 'Open' button) then the menu display should change to 'Open, No Pack'. If the pawl arm is operated then the menu display should change to 'Open, Pack Present'; if the pack door switch is operated as well then the menu display should change to 'Closed, Pack Present'.
Bar Code Scanner:	The value of this menu item shows the value of last barcode read. If 'XXX' appears, the sensor is faulty. The 'Enable' button initiates the read sequence of the barcode reader. The 'Disable' button stops this process. Note: the barcode laser will deactivate on a successful read operation even if 'Disable' has not been pressed.
Bar Code Alignment:	This menu item activates the barcode reader laser to emit a wide-angle beam that can reach the moulded alignment marks inside the C600. These allow adjustment if required without the door or a pack in position. The 'Test' button activates the laser and does not deactivate it after a successful read operation. The 'Stop' button deactivates the barcode laser. If 'XXX' appears, the sensor is faulty.

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7.2 Technician Settings

Pack Door Cycle Time:	This menu item is used for diagnosis of the pack door motor. Selecting 'Open' will initiate one full rotation of the pack door motor and display the cycle time in centi-seconds.
Pack Door Motor:	This menu item shows the current status of the door latch switch. The 'On' button will activate the pack door motor. The latch switch value should change between 'Not Home' and 'Home' as the motor rotates. The 'Off' button ceases the rotation at the current location.
Handling Mech:	This menu can be used to test the operation of the brewer's pack handling mech and its top and bottom datum switches. The menu displays the current state of the handling mech's top datum switch input ('Top' or 'Not Top') and the current state of its bottom datum switch input ('Bottom' or 'Not Bottom'). Pressing the 'Down' button will run the handling mech motor until the bottom datum switch is operated, piercing any pack in the process. At this point the menu display should change to 'Bottom, Not Top'. If the handling mech is already at the bottom of its travel then the 'Down' button will have no effect. Pressing the 'Up' button will run the handling mech motor until the top datum switch is operated, ejecting any pack in the process. At this point the menu display should change back to 'Not Bottom, Top'. If the handling mech is already at the top of its travel then the 'Up' button will have no effect. Note: 1) The handling mech will run quite safely without a pack being present. 2) If 'Not Bottom, Not Top' is displayed permanently then this indicates that either the handling mech has jammed with neither datum switch made or the connections from the datum switches to the main control board have broken. 3) If 'Bottom, Top' is displayed permanently (i.e. both datum switches appear to be made) then this indicates that one or both datum switches are not positioned correctly.

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7.2 Technician Settings

Reverse H-Mech Motor:	The 'Nudge' button moves the handling mechanism motor in reverse for a very short period, but only when the mechanism is between the Home positions. This is intended to assist in the clearance of pack jams.
Inject Time /ms:	This is an indicator showing the time taken for the handling mechanism to move from the 'Home' position to the 'Bottom' position.
Eject Time /ms:	This is an indicator showing the time taken for the handling mechanism to move from the 'Bottom' position to the 'Home' position.
H-Mech IJ Calibration:	This menu item allows the self-calibration of the handling mechanism motor speed. The 'Start' button will run the handling mechanism for a short period and stop automatically once the calibration is complete.
H-Mech IJ Cycle Time:	This is an indicator showing the handling mechanism Inject cycle time. Pressing 'Start' will cause the handling mechanism to run continuously at injection speed, updating the cycle time every cycle.
H-Mech EJ Calibration:	This menu item allows the self-calibration of the handling mechanism motor speed. The 'Start' button will run the handling mechanism for a short period and stop automatically once the calibration is complete.
H-Mech EJ Cycle Time:	This is an indicator showing the handling mechanism Eject cycle time. Pressing 'Start' will cause the handling mechanism to run continuously at injection speed, updating the cycle time every cycle.
Pack Bin Detector:	This is an indicator showing the status of the pack bin detection switch. The value should read 'Bin Present' when the switch is actuated and 'Bin Not Present' when it is in its natural position.
Drip Tray Detector:	This is an indicator showing the status of the drip tray detection reed switch. The reed switch is activated by a magnet located in the drip tray upper. The value should read 'Tray Present' when the drip tray is installed and 'Tray Not Present' when it is removed.

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7.2 Technician Settings

Cup Stand Calib State:	This indicator shows the current state of the cup stand motor calibration. The 'Up' and 'Down' buttons will move the cup stand. The possible values are: 'Uncalibrated', 'Part Calibrated (Btm)', 'Part Calibrated (Top)', 'Calibrated (at Top)', 'Calibrated (at Home)'
Cup Stand Calib Time:	This indicates the time in centi-seconds of the most recent cup calibration. Pressing 'Start' triggers the cup-stand to calibrate.
Cupstand Encoder Type:	This indicator shows the type of cup stand motor encoder in use on this brewer. The menu may read are: 'Optical' 'Capacitive' 'Waiting to Detect' 'Failed to Detect'
Cup Stand Encoder:	This menu item shows the current cup stand encoder value. The 'Up' and 'Down' buttons will move the cup stand in the directions indicated. While moving, the cup stand encoder value should change with a higher value indicating a higher position.

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Cup Sense Inputs /mV:	This menu displays, in millivolts, the voltages being read from the 3 cup sensor circuits in the order Top, Middle, Bottom. The values will depend on the calibration state of the circuits and on the cleanliness of the lenses. With nothing blocking the beams, the voltages should all be in the range 0 to 600 mV (some signal noise and some drift in each voltage is expected). If any of the voltages are above 600 mV then first try cleaning both cup sensor lenses; if that fails to lower the voltages below 600 mV then re-calibrate the cup sensors using the 'Cup Sense Calibration:' menu. With an opaque object blocking a beam, its voltage should be in the range 2200 to 2700 mV. If any of the voltages are above 2700 mV then there is a fault in the cup sensor circuits. If any of the voltages are below 2200 mV then remove the opaque object, clean both cup sensor lenses and then re-calibrate the cup sensors using the 'Cup Sense Calibration:' menu. The 'On' and 'Off' buttons help diagnose faults with the sensor circuits by enabling and disabling the power and clock to the transmitters. After pressing 'Off', the 3 voltages should all change to 0 mV; if they do not then there is a fault in the cup sensor circuits.
Cup Detectors:	This menu item gives a visual indication of the status of each cup detect sensor. A 'Y' indicates a cup is detected, a 'N' indicates no cup detected, and a 'Y?' indicates an illogical cup detection (e.g. middle sensor but not lower sensor).
Cup Detection System:	This menu indicates the overall view of the cup detection system that is used when vending drinks. It takes into account not just the three cup sensors but also the presence of the drip tray, the state of cup stand calibration and the height of the cup stand.

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Cup Stand Motor:	To help diagnose problems with the motor, gearbox and encoder, this menu allows the cup stand motor to run without any of the usual dependencies on cup sensors, cup stand encoder and drip tray detector. There is not even a requirement to mount the motor in the machine. In each direction of travel, the motor times out and then reverses direction.
Pack Heater:	This item shows the current status of the pack heater. The 'On' button will activate the pack heater element and pack fan. The pack heater will deactivate after 19s but the pack fan will continue running for a period to cool the assembly. The 'Off' button will deactivate the heater but the pack fan will continue to run.
Priming Valve:	This menu opens the Priming Valve allowing water to return to the cold tank.
Injector Valve:	This menu opens the Injector Valve allowing water towards the injector path.
Jet Valve:	This menu opens the Jet Valve allowing water towards the jet path.
Water Pump Motor:	This menu allows the water pump to be turned 'On' and 'Off' manually.
Air Pump Motor:	This menu allows the air pump to be turned 'On' and 'Off' manually.
Pressure Sensor /psi:	This is an indicator of the current Pressure sensor reading in PSI. If 'XXX' appears, the sensor is faulty.
Pressure Switch:	This menu item shows the current status of the pressure switch. The value may read: 'Pressure OK' 'Overpressure' 'Pressure Unknown' 'Error (circuit fault)' The 'Test' button opens the door, lowers the injector and activates the air pump. Blocking the injector should result in an overpressure and the air pump will deactivate.
Flow Meter Status:	This menu item shows the status of the flow meter. The value may read: 'Meter Untested' 'Meter Faulty' 'Meter Working OK'

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Pump Metered Water:	Opens the injector valve and starts the water pump. The valve and water pump are automatically deactivated after 250 pulses have been received from the flow meter.
Internal Cold Tank:	This menu displays the current status of the cold tank float switch. The possible values are: 'Tank Level High' 'Tank Level Low' 'Tank Level Faulty' 'Tank Level Unknown' If the float is detected as low and the fill timeout has been reached, the 'Retry' menu item appears. This will open the mains inlet valve until either the float is detected in the upper position or the fill timeout is reached.
External Water Tank:	Shows the current fill status of the water tank in the Flavia REFILL module.
Chiller Air Pump:	This menu allows manual control of the Air pump in the Flavia CHILL module. Note: Running the air pump has the side-effect of resetting the chiller's water metering system. Any cold water left in the system after testing should be expelled via this menu before exiting the menus and vending drinks. Failure to do so will result in a shorter volume for the next cold drink, and abnormal pack opening and temperature of the first vend of any type of drink.

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Chiller Water Pump:	This menu allows manual control of the Water pump in the Flavia CHILL module. Note: 1) After approx. 10 fl. oz of water has been dispensed, the chiller will automatically stop pumping and the menu value will change back to 'Motor Off'; the chiller will not pump water again until its water metering system has been reset by running the chiller air pump from the 'Chiller Air Pump' menu described above. 2) Attempting to pump chiller water with all C600 manifold valves off will initially trigger the warning message: 'ALL VALVES ARE CLOSED! Press ON again if sure'. Only proceed for the purpose of testing the chiller's PRV and leak detection system, which should then produce chiller "Error 1".
Chiller Status:	This menu displays the current status of the Flavia CHILL module. The menu may read: "Temperature <= 3 degC", "Temperature 4-12 degC", "Temperature 13-20 degC", "Water Pump Running", "Water Level Low", "Error (check LED)", "Temperature > 20 degC", "Module Turned Off" Note: Vending of cold drinks will only be allowed when the Chiller Status is either 'Temperature <= 3 degC' or 'Temperature 4-12 degC'. In all other states the cold drink selections on the UI will be greyed-out and unselectable.
PSU 5V Supply:	Displays the status of the Main PCBA 5V Supply
PSU 24V Supply:	Displays the status of the Main PCBA 24V Supply. The menu will read a combination of the current state of the Supply: 'Turned off' or 'Turned On' And the sensed level of the 24V Supply: 'No 24V' or '24V OK'. e.g. 'Turned on, 24V OK'

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Pack Fan Status:	This menu allows manual control and displays the status of the Pack fan. The menu may read: 'Fan Off' 'Starting Up' 'Retrying' 'Fan OK' 'Fan Error (Slow)' 'Fan Error (Stalled)'
Pack Fan Speed (Hz):	Shows the frequency of the speed signal received from the pack fan. The pack fan produces a pulse twice per revolution. This menu also allows manual control of the pack fan.
Pack Heater:	Shows the status of the pack heater. This menu also allows manual control of the pack heater. Note: When the pack heater is activated the pack fan will also activate. Once the heater is turned off, either automatically or manually, the pack fan will continue to run to cool the system down.
Hot Tank Test Mode:	This menu can be used to test the operation of the two hot tank heater elements. The menu displays the current mode that the heater test is in ('All Heaters Off', 'Pre Tank Heater On', 'Main Tank Heater On'). Once a tank heater test is selected, the heater element in the relevant tank will be energised for 2 seconds before reverting to 'All Heaters Off'.

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Pre Tank Status: Main Tank Status:	These two menus display the current status of the Pre and Main Hot Tanks. Each tank could have the following status • 'Low Temperature' – the tank's thermistor is functioning ok, the tank is under normal heating control but the water temperature is below 75°C / 167°F. • 'Under Temperature' – the tank's thermistor is functioning ok, the tank is under normal heating control but the <i>water temperature is below</i> the vending threshold (90°C / 194°F). • 'Temperature OK' – the tank's thermistor is functioning ok, the tank is under normal heating control and the water temperature is within the allowed vending range (90°C - 94°C / 194°F - 201°F). • 'Over Temperature' – the tank's thermistor is functioning ok, the tank is under normal heating control but the <i>water temperature is above</i> the vending threshold (94°C / 201°F). • 'Empty (unprimed)' – the tank has <i>not been fully primed</i> so heating control has been disabled. The other tank will also have this status. • 'Heater Disabled' – the tank has been primed but heating control has been disabled by the 'Hot Tank Control:' menu. The other tank will also have this status. • 'Error (thermistor O/C)' – the tank's <i>thermistor appears to be open-circuit</i> (e.g. broken wire, disconnected loom) so heating control has been disabled. • 'Error (thermistor S/C)' – the tank's <i>thermistor appears to be short-circuited</i> (e.g. wires crushed together or to earthed metalwork) so heating control has been disabled.
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Hot Tank Control:	This menu displays the current heating control state of the brewer's Hot Tanks and allows the heating to be 'Enabled' or 'Disabled'. The option 'Prime' will begin the sequence to fill the Hot Tanks from the cold water tank. This option will have no effect if the heaters are enabled. The 'Disable' option will deactivate the heaters in preparation for a hot tank drain. This option will have no effect if the heaters are already disabled. The Hot Tank status can be one of: 'Disabled (Tanks Empty)' meaning that hot tank heating control is turned off because the hot tanks have not yet been primed (filled with water). 'Disabled (Priming...)' meaning the hot tanks are being filled with water but the heaters remain disabled until this has been completed without error. 'Enabled (Tanks Primed)' meaning the hot tanks are full and normal hot tank heating control is in progress. 'Disabled (OK to Drain)'. Meaning the heaters are switched off and the system is ready for draining.
Hot Tank Flush:	Flushing the hot water system is recommended when the brewer is new "out-of-box" or following any period of prolonged dormancy. The menu shows a summary of the Hot Tank status which can be one of: 'Tanks Empty' 'Tanks Primed' 'Tanks Flushing...' 'Tanks Priming...' 'Tanks Filled' 'Tanks Draining...' Pressing the 'Flush' button will initiate a system flush through either the Jet or Injector path. Once complete, pressing the 'Flush' button again will bring a flush of the other path. The path flushed will alternate with each press. It is recommended to run the flush cycle twice to cover both water paths. The 'Stop' button interrupts the current flushing cycle. Note that the flushing cycle is intermittent to allow time for the cold tank to refill and will pause and restart automatically whilst the screen displays "Tanks Flushing". The flush cycle will automatically stop once it is complete.

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Hot Tank Drain:	This menu shows a summary of the Hot Tank status as in the 'Hot Tank Flush' menu above. The 'Drain' button will begin the active cycle to purge water from the Hot Tanks through the hot tank drain hose. Please ensure the Cold Tank has been drained prior to starting a Hot Tank Drain. Pressing 'Stop' will abort the drain cycle if it is running.
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7.2.3 Modules

FLAVIA® CHILL	This menu is used to select if a Flavia CHILL is connected to the brewer. The options are: 'Not Fitted' 'Fitted with 2' Water Pipe' 'Fitted with 7' Water Pipe'
FLAVIA® REFILL	This menu is used to select if a Flavia REFILL is connected to the brewer. The options are: 'Fitted' 'Not Fitted'

7.2.4 Resets

Restore Defaults	Restores the default brewer in all contextual menus. Note: any changes made in the Engineering Menu will not be affected by this reset.
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7.2.5 Operating Modes

Machine Mode	Hardware Check – For factory use only Configuration Setup – For factory use only End of Line – For factory use only Installation – Restores the brewer to the initial installation mode. This includes restoring the default settings as described in 7.2.3 Normal – The normal operating mode of the brewer
Serial No. Validation	If checked (normal) the brewer validates the serial number at boot-up. This may be disabled if there was a problem with brewer registration during a SBC or head-faceplate replacement. Serial number validation can only be enabled, it cannot be disabled. Please see Section 5.6.4.3 for more information

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7.2.6 Flash/Clone

Software Up To Date/ Software Available X.X.X.X	Select to retrieve the latest software and install this onto the brewer. Once selected a confirmation dialogue will be shown.
Firmware Up To Date / Firmware Available X.XX	Select to retrieve the latest brewer controller firmware and install this onto the brewer. Once selected a confirmation dialogue will be shown.
Clone From Key	This menu item allows the restoration of all site-specific settings from a USB key to speed up installations of multiple brewers on the same site. Insert USB Key – No USB Key detected No Clone File Present – USB Key detected but clone file is not present Clone File Present – USB Key detected and file present, select this item to copy the clone from the USB Key to the brewer Cloned From Key – Clone file successfully copied to the brewer
Clone To Key	This menu item allows the exporting of all site-specific settings to a USB key for backup or deployment to other machines. Insert USB Key – No USB Key detected Enabled – USB Key detected and ready to clone Cloned To Key – Clone created successfully
Copy Database	This item allows the brewer's vend and error history to be manually downloaded to a USB key in the event of a IoT connectivity failure. Insert USB Key – No USB Key detected Enabled – USB Key detected and ready to copy Database Copied – Database copied successfully
Eject USB Key	Dismounts the USB Key ready for removal. This menu item is only shown if a USB Key is detected.

The software and firmware are always supplied as a matching pair. When the software is updated the brewer may update the firmware automatically as part of the process.

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7.3 Error Messages and Events

7.3.1 Error Messages

Faults that the control system can detect will be indicated on the LCD display. If in Vend Mode then a 'Please Call Operator' screen will be displayed along with an error code and a description; if in the menu system then just an error message (i.e. a short description of the fault) will be displayed.

Each detectable fault has a unique **error code number** assigned to it; this is a three-digit number starting with a '3'.

The error code numbers are chosen so that: (a) common or severe errors have memorable numbers, (b) if the digits are mixed up then the number is still unique and the mix-up should be obvious and (c) categorised so that the fault area can be identified from the middle digit.

The error code number is displayed on the 'Please Call Operator' screen as 'ERROR Xxx'.

If the menu system is in use when the fault is detected then the current menu entry will be temporarily overwritten by the error message; the original menu display can be restored by pressing the 'Continue' button. (And likewise if the menu system is entered *after* an error has been reported in Vend Mode).

For some errors, the act of simply entering and exiting Menu Mode will clear the 'Please Call Operator' screen (note that this will not necessarily clear the actual fault). For other errors, the 'Please Call Operator' screen will persist when Vend Mode is re-entered.

The menus mentioned in this section are described in full in Section 7.2 or Section 7.3.2.

The term “main control board” refers to the C600 Brewer Controller PCBA.

The error codes, the menu system error messages, their causes and appropriate repair actions are listed in the following table:

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ERROR CODE	ERROR MESSAGE	CAUSE	ACTION
302	COUNTERS ZEROED	Should only appear if the PCO and Event counters have been zeroed in the 'Manage PCO/Event Log:' menu.	This is a warning / feedback message rather than an error message. No action is required.
305	COLD TANK NOT FULL	Should only appear if attempting to start pumping in the 'Pump Metered Water:' menu when the cold tank has not yet filled up to the level of the float switch.	This is a warning / feedback message rather than an error message. Wait for the cold tank to fill; if there has been a filling timeout then a filling "retry" would be automatically triggered by the menu action.
321	PACK INJECT TIMEOUT	Filter pack inject failed – a datum switch was not made within the expected time from starting the handling mech motor.	Manually open the pack door and use the 'Reverse H-Mech Motor:' menu to free the pack nozzle. Clear the pack and then exit Menu Mode to reset the mechanism. Use the 'Handling Mech:' menu to help diagnose switch faults. Check that the pack flap is free-moving and can correctly align pack nozzles with the Injector. Check the handling mech loom to see if it is unplugged or broken or shorted to itself. A faulty / disconnected motor may also be the cause.
325	H-MECH SWITCH FAULT	The handling mech datum switches are not in a valid state - both switches are made at the same time. This condition inhibits the normal "inject" and "eject" operations which use the datum switches to determine where and when to stop the motor.	Use the 'Handling Mech:' menu and the 'H-Mech IJ Cycle Time:' menu to see if both switches are made permanently or only intermittently. Check the mechanism for a faulty / dislocated barrel cam switch. Check the handling mech loom to see if it is shorted to itself.

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ERROR CODE	ERROR MESSAGE	CAUSE	ACTION
326	H-MECH CCT OVERLOAD	The circuit to the handling mech motor has drawn too much current and has been shut down.	Check the handling mech motor wiring for short-circuits to earth/0V or to itself. A faulty motor may also be the cause. Use the 'Handling Mech:' menu to check if the fault is permanent or intermittent. Note that it is possible for the fault to occur in only one direction of rotation.
328	PACK EJECT TIMEOUT	Filter pack eject failed - a datum switch was not made within the expected time from starting the handling mech motor.	Manually open the pack door and check if the pack bin is overflowing and obstructing the pack ejection path. Use the 'Reverse H-Mech Motor:' menu to un-jam the mechanism, then use the 'Handling Mech:' 'Up' menu to try to eject the pack and reset the mechanism. Repeat if necessary. Use the 'Handling Mech:' menu to help diagnose switch faults. Check the handling mech loom to see if it is unplugged or broken or shorted to itself. A faulty / disconnected motor may also be the cause.
333	ESPRESSO COMMS FAULT	Communications between the main control board and the espresso control board have failed, even after several attempts to reset and re-initialise the espresso board. Note: No espresso board is available for C600	This is not expected in a C600 brewer. A configuration setting has become corrupted. Check if this error is accompanied by Error 395 'NVRAM CORRUPT' and / or the 'NVRAM State:' menu has become visible. If the menu is visible then use the 'Reset' option, otherwise replace the main control board.

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ERROR CODE	ERROR MESSAGE	CAUSE	ACTION
341	CUP STAND FAULT	The cup stand motor did not reach its target encoder position within the expected time from starting the motor, even after several retries. This includes failures to detect the upper half of the cup stand at the appropriate point and obstructions detected after calibration has started.	Check for obvious obstructions above or below the moving parts of the cup stand (e.g. an undetected cup or the upper half being disengaged from the pack bin guide channel). Check if the drip tray is fully engaged with the chassis and the drive peg. Remove the drip tray and make sure that the cup stand mechanism returns to the fully 'collapsed' position. Clean the cup stand sections, the drip tray and the front of the bin. Use the 'Cup Stand Calib Time:' menu to check for full, free and consistent movement of the cup stand mechanism during calibration.
343	CUP STAND ENCODER FAULT	The positional encoder on the cup stand motor is not responding when the motor is being run. (It is not caused by a jammed cup stand mechanism.)	A faulty / disconnected cup stand encoder is one possible cause – check this loom to see if it is unplugged or broken or shorted to itself. A faulty / disconnected cup stand motor is another possible cause. Use the 'Cup Stand Motor:' menu to see if the motor runs ok and to see if the encoder fault is permanent or intermittent.
344	CUP STAND CCT OVERLOAD	The circuit to the cup stand motor has drawn too much current and has been shut down.	Check the cup stand motor wiring for short-circuits to earth/OV or to itself. A faulty motor may also be the cause. Use the 'Cup Stand Encoder:' menu to check if the fault is permanent or intermittent. Note that it is possible for the fault to occur in only one direction of travel.
351	VALVE DISCONNECTED	One of the solenoid dispense valves appears to be disconnected.	Check the wiring to the dispense valves for unplugged connectors or broken wires or crimps. A faulty valve coil may also be the cause.

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ERROR CODE	ERROR MESSAGE	CAUSE	ACTION
353	VALVE OVERLOAD	One of the solenoid dispense valve circuits has drawn too much current and has been shut down.	Check the wiring to the dispense valves for short-circuits to earth/0V or to itself. A faulty valve coil may also be the cause.
355	REFILL CIRCUIT OVERLOAD	The circuit to the water pump relay in the FLAVIA® REFILL water tank module has drawn too much current and has been shut down.	Check the wiring to and inside the FLAVIA® REFILL water tank module for short-circuits to earth/0V or to itself. A faulty water pump relay coil may also be the cause.
360	PACK DOOR TIMEOUT	The pack door opening cam has not left and then reached 'Home' within 3 secs.	Remove the pack door and check if the latch arm can be moved up and down freely. Check for a build-up of ingredients around the door latch. Use the 'Pack Door xxx:' menus to check if the latch arm is moving down and up freely when the motor runs and to see if the 'Pack Door Cycle Time' is high (more than 110 centi-seconds). A faulty motor or a faulty 'Home' switch or a broken connection may also be the cause.
363	PACK DOOR OVERLOAD	The pack door motor circuit has drawn too much current (even after several retries) and has been shut down.	Check the pack door motor wiring for short-circuits to earth/0V or to itself. A faulty motor may also be the cause.
371	SYSTEM OVER-PRESSURE	The filter pack system pressure switch has activated (even after several retries) and the pumps have been shut down.	Check for (and clean out) any limescale in the Injector. Check the moulded water pipe above the Injector for kinks or lime scale. Check that the pack flap is free-moving and can correctly align pack nozzles with the Injector. Check if an espresso pack has been forced into the pack rail.

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ERROR CODE	ERROR MESSAGE	CAUSE	ACTION
373	PUMP OVERLOAD	The motor circuit for either the water pump or the air pump has drawn excessive current (even after several retries) and has been shut down.	Use the 'Water Pump Motor:' menu and 'Air Pump Motor:' to check if the fault is permanent or intermittent. Check that the pump motor shaft is running freely. Check the pump motor wiring for short-circuits to earth/0V or to itself. A faulty pump motor or a seized pump head may also be the cause (though less likely with the air pump).
374	PACK FAN FAULT (STALL)	The pack fan has stalled (even after several retries).	Check the operation of the fan using the 'Pack Fan Status:' and 'Pack Fan Speed (Hz):' menus, and see if the fault is permanent or intermittent. Check the 'Event Y Count:' menu value for previous history of recoverable pack fan stalls. Check the pack fan loom to see if it is unplugged or broken or shorted to itself. A faulty fan motor or obstructed fan blades may also be the cause.
375	PACK FAN FAULT (SLOW)	The pack fan is running abnormally slow. This is most likely to be caused by a physical restriction or fault with the fan itself.	Check the operation of the fan using the 'Pack Fan Status:' and 'Pack Fan Speed (Hz):' menus. Check for obstructions or mess around the fan blades. The fault may not be easily reproducible; it could be temperature related.

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ERROR CODE	ERROR MESSAGE	CAUSE	ACTION
377	FLOW METER TIMEOUT	The flow meter for the water pump is not responding suitably when the water pump is running. This is caused either by the flow meter and pump being starved of water or by a faulty / disconnected flow meter or water pump.	Check if the cold tank access lid is fitted and fully seated. Check if the water filter in the cold tank is fitted and fully seated. Check that the float switch is free-moving and that the cold tank is full to the level of the float switch. Check if the water pump can be heard running. Use the 'Flow Meter Status:' and 'Water Pump Motor:' menus to check if the fault is permanent or intermittent. Check the 'Event Z Count:' menu value for previous history of heavily restricted flow. Check the flow meter loom to see if it is unplugged or broken or shorted to itself or corroded. A faulty / blocked flow meter may be the cause. Check the water pump wiring to see if it is unplugged or broken. A faulty / seized water pump may be the cause.
379	HOT TANK DRAIN BLOCKED	The hot tank draining process has been aborted due to high pressure detected in the manifold – probably because the hot tank drain pipe is still / now bunged. (The abort process relieves the pressure in the system, via the Priming valve, to reduce the danger when the blockage is removed.)	Check if the hot tank drain pipe is still fitted in its chassis kink features or is still bunged or has become kinked. If the draining process was started intentionally then it can be re-started from any stage with the 'Hot Tank Drain:' menu.

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ERROR CODE	ERROR MESSAGE	CAUSE	ACTION
383	24V POWER TRIPPED	The Protected 24V power has drawn too much current and has been shut down. There is a problem with one of the devices or wires that connect to the Protected 24V power supply on the main control board. The software has tried resetting the electronic trip circuit but has given up as there is a persistent fault.	Check the 'PSU 24V Supply:' menu to see if the 24V has recovered - this would indicate an intermittent fault, as would the 'Event N Count:' menu value. Check for obvious short circuits and failed devices. Unplug the looms on the main control board that carry 24V. Replace them one at a time, powering off/on in between, to try to isolate the source of the fault. If necessary, refine the search by unplugging the loom at each device rather than at the board. Note that the various motors are themselves unlikely to cause this error due to the protection in their drive circuits and the detection and reporting of such faults (see the 'xxx OVERLOAD' errors).
391	C600 PCBA INCOMPATIBLE	The design-level of the main control board (C600 PCBA) is not compatible with the functionality of the machine that the software is expecting.	If the error is seen after replacing and re-programming the main control board on a FLAVIA® C600 then the replacement board is most likely of an old design and must be replaced with a newer version that is capable of controlling the latest design of C600 - check the artwork issue no. in the silk-screen. A faulty main control board may also be the cause.
393	NVRAM WRITE FAILED	An attempt to write to the NVRAM has failed - even after several retries. Possibly due to an electrically noisy environment or water damage to the main control board or the NVRAM device failing.	Replace the main control board.

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ERROR CODE	ERROR MESSAGE	CAUSE	ACTION
395	NVRAM CORRUPT	The configuration data stored in NVRAM (Non-Volatile RAM) does not match its checksum. Detected only at power-up. Possibly due to (a) an electrically noisy environment, (b) the NVRAM device failing, (c) re-programming with an older version of software. (Note: in contrast, any corruption of the PCO/Event counters is “fixed” automatically.)	Check the configuration settings; if they are garbage then use the 'Reset' option in the 'NVRAM State:' menu described in Section 7.2. If the settings appear uncorrupted then use the 'Fix' option in the 'NVRAM State:' menu. If this fails to clear the error then replace the main control board.
399	NVRAM FAULTY	NVRAM device failure detected at power-up. Possibly due to a fault / damage on the main control board or the NVRAM device itself failing.	Replace the main control board.

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7.3 Error Messages and Events

7.3.2 Event Log Mode

The software logs the number of occurrences of certain notable errors and events; these counters are stored in non-volatile memory (NVRAM) and can be viewed in the 'Event Log Mode' menus. This section describes these menus and the circumstances in which each error/event is logged.

To enter 'Event Log Mode':

- Enter Menu Mode (described in Section 7.2) and then on the 'Settings' screen enter the password **4 3 3 3** on the displayed keypad and press **OK**.
- The LCD will then change to show the 'Event Log' menus.
- The 'Event Log Mode' menus can be navigated in the same way as the 'Engineering Mode' menus.

'**PCO X Count:**' menus are for certain '**P**lease **C**all **O**perator' errors that result from actions initiated by the consumer in Vend Mode. Note that some errors cannot be logged (i.e. NVRAM-related errors) and some are not worth logging (e.g. warnings).

'**Event Y Count:**' menus are for interesting and notable "events" that occur at any time (by their definition, some will only occur in Vend Mode whereas others are perfectly valid in any mode). Some events simply monitor how the brewer is being used whereas others indicate "soft-faults" (i.e. detectable malfunctions that do not prevent the brewer from continuing to vend). These counters can indicate the need for preventative maintenance so that more serious faults do not develop.

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Note:

- 1) These “counter” menus appear in the order listed in the table below; they are followed by a number of configuration and control menus (described below).
- 2) To avoid confusion, the ‘**PCO Counts**’ are numbered whereas the ‘**Event Counts**’ start with a letter.
- 3) The “**severity**” of an error or event in the table below gives an indication of the relative importance of any non-zero counter values seen; the higher the number, the higher the severity. A large number of relatively unimportant errors/events should be investigated in a similar way to a few occurrences of very severe errors/events.
- 4) Setting the ‘**Vend Mode:**’ menu in ‘Engineering Mode’ to ‘**Test**’ will suppress the logging of errors and events until the end of the next vend.
- 5) The maximum value for each counter is 65,000. Once a counter reaches its maximum value, it will stay at that value until explicitly zeroed by the ‘**Zero all Counters**’ menu command described below.

‘**Last Event:**’ read-only menu gives the ID of the last “event” reported by the control system (the possible events are listed in the table below). This event is stored in NVRAM.

‘**Last Error:**’ read-only menu gives the last error message reported by the control system (the possible errors are listed in the table above). It does not record warning / feedback messages. This error is stored in NVRAM.

‘**Cause of System Reset:**’ read-only menu gives the cause of the last microprocessor system reset on the main control board. Possible reset causes are:

- ‘**Power-Up**’ (this is the normal cause of reset)
- ‘**Brown-Out**’ (see ‘**Event S**’ and Note 15 below)
- ‘**Internal Watchdog**’ (see ‘**Event O**’ and Note 11 below)
- ‘**External Watchdog**’ (see ‘**Event P**’ and Note 11 below)
- ‘**Flash Reprogram**’ (if the board’s software has been reprogrammed)

‘**Vends Since Power-Up:**’ read-only menu gives the number of packs and hot water dispenses vended since the brewer was last powered-up.

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‘Manage PCO/Event Log: Zero all Counters?’ command menu allows the PCO and Event counters to be reset to zero; when the **‘Start’** button is pressed, **all** the counters are zeroed and the **‘COUNTERS ZEROED’** warning message is displayed as feedback. This menu can be used to simplify the monitoring of errors and events.

‘Total Cups @Log Zero’ read-only menu gives the value of the **‘Total Cups:’** audit counter when the above **‘Zero all Counters’** menu command was last used. By comparing this value to the current value of the **‘Total Cups:’** menu, the PCO and Event counters values can be put into context. This value is stored in NVRAM.

‘PCO/Event Logging: Enabled/Disabled’ configuration menu allows the logging of errors and events to be temporarily or permanently disabled (e.g. to avoid logging false counts whilst experimenting with a brewer). If logging is **disabled** then the “counter” menus will not be displayed.

‘Machine ID:’ configuration menu allows a unique identifying number (between 0 and 65,000) to be stored in the brewer; this menu is transmitted as part of the **‘Machine Audit’** (see below) so that the data can subsequently be matched to a particular brewer. This ID number has no other significance.

‘Host Serial Number:’ read-only menu gives the serial number of the Brewer that was sent from the UI-Host. Normally, this will match the serial number seen on the UI-Host’s initial **‘Settings’** screen. This menu is transmitted as part of the **‘Machine Audit’** (see below) so that the data can subsequently be matched to a particular brewer.

‘Machine Audit: Send all Data ?’ command menu allows the PCO and Event counters to be transmitted out of the serial port (along with all the audit and configuration data). The data takes around 18 seconds to transmit; it is sent "in the background" so the menus will carry on operating as normal during the transmission. A special cable is required to connect the brewer to a PC.

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Menu Title of Counter	Description	Severity
PCO 1 Count:	Number of vends aborted by 'PACK INJECT TIMEOUT' PCO errors	4
PCO 2 Count:	Number of vends aborted by 'PACK EJECT TIMEOUT' PCO errors	2
PCO 3 Count:	Number of 'PACK DOOR OVERLOAD' PCO errors caused at the start of a vend (see Note 4)	10
PCO 4 Count:	Number of 'PACK DOOR TIMEOUT' PCO errors caused at the start of a vend (see Note 4)	6
PCO 5 Count:	Number of vends aborted by 'SYSTEM OVER-PRESSURE' PCO errors (see Note 1)	7
PCO 6 Count:	Number of 'CUP STAND TIMEOUT' PCO errors	2
PCO 7 Count:	Number of 'CUPSTAND ENCODER FAULT' PCO errors	10
PCO 8 Count:	Number of vends aborted by 'PUMP OVERLOAD' PCO errors	10
PCO 9 Count:	Number of '24V POWER TRIPPED' PCO errors	10
PCO 10 Count:	Number of 'VALVE DISCONNECTED' PCO errors	10
PCO 11 Count:	Number of 'VALVE OVERLOAD' PCO errors	10
PCO 12 Count:	Number of 'H-MECH SWITCH FAULT' PCO errors	10
PCO 13 Count:	Number of 'PACK FAN FAULT (STALL)' PCO errors (see Note 21)	10
PCO 14 Count:	Number of 'PACK FAN FAULT (SLOW)' PCO errors (see Note 20)	9
PCO 30 Count:	Number of 'FLOW METER TIMEOUT' PCO errors	8
PCO 31 Count:	Number of 'REFILL CIRCUIT OVERLOAD' PCO errors	10
PCO 32 Count:	Number of 'H-MECH CCT OVERLOAD' PCO errors	10
PCO 33 Count:	Number of 'CUP STAND CCT OVERLOAD' PCO errors	10
Event A Count:	Number of times a pack was detected <i>after</i> the pack door finished opening at the start of a vend (see Note 4)	2
Event B Count:	Number of vends initiated by the pack door switch <i>preceding</i> the pack detect switch (see Note 8)	1

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Menu Title of Counter	Description	Severity
Event C Count:	Number of times vends were paused around pack injection time by the pack door switch (see Note 7)	5
Event D Count:	Number of times vends were paused around pack injection time by the pack detect switch (see Note 7)	6
Event E Count:	Number of times vends of any type were paused at any stage by entering Menu Mode	1
Event F Count:	Number of vends aborted by "pack absent" timeouts (see Note 5)	7
Event G Count:	Number of times the pack injector overran its top home position at the end of a pack ejection cycle (see Note 2)	6
Event H Count:	Number of times the pawl arm did not return home at the end of a pack ejection cycle (see Note 10)	4
Event I Count:	Number of times vended water shots were interrupted (recoverably) by the filter pack system over-pressure switch (see Note 1)	5
Event J Count:	Number of times the "packs in bin" counter has been zeroed (see Note 3)	0
Event K Count:	Number of times the brewer has been powered up	0
Event L Count:	Number of times the pack door switch never opened during the pack door opening cycle (see Note 4)	5
Event M Count:	Number of times the pawl arm reset cycle failed (see Note 9)	5
Event N Count:	Number of recoverable over-current trips on the Protected 24V supply on the main control board	8
Event O Count:	Number of times the H8 microprocessor on the main control board has been reset by its internal watchdog (see Note 11)	9
Event P Count:	Number of times the H8 microprocessor on the main control board has been reset by its external watchdog (see Note 11)	8
Event Q Count:	Number of times the Barcode scan engine has been software-reinitialised to recover after an error was detected (see Note 12)	7
Event R Count:	Number of times the Barcode scan engine has failed to respond to serial communications	9

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Menu Title of Counter	Description	Severity
Event S Count:	Number of times the H8 microprocessor on the main control board has been reset by a power supply "brown-out" (see Note 15)	3
Event T Count:	Number of times a pack's barcode could not be read at the start of its vend (see Note 13)	6
Event U Count:	Number of one-pack or two-pack vends aborted by a "cup absent" timeout at any point not covered by Event V (see Notes 16 and 17)	1
Event V Count:	Number of one-pack or two-pack vends aborted due to the cup being removed during the "jet water phase" of either pack (see Notes 16 and 18)	1
Event W Count:	Number of one-pack or two-pack vends aborted by the consumer cancelling the vend during the brewing process of either pack	0
Event X Count:	Number of two-pack vends aborted by the consumer cancelling the vend at the mid-point instead of going on to vend the 2 nd pack	0
Event Y Count:	Number of recoverable pack fan stalls (see Note 21)	5
Event Z Count:	Number of times vended water shots have had heavily restricted flow as detected by the flow meter (see Note 22)	6
Event A1 Count:	Number of times one of the water pump calibration figures reached its maximum allowed value	9
Event B1 Count:	Number of user-recoverable Cup Stand faults	2
Event E1 Count:	Number of recoverable pack <i>Eject</i> timeout faults	5
Event H1 Count:	Number of times vended <i>cold</i> water shots were aborted by the FLAVIA® CHILL module entering a fault state	7
Event I1 Count:	Number of recoverable pack <i>Inject</i> timeout faults	5
Event J1 Count:	Number of times the cold tank has failed to fill sufficiently in the time allowed	2
Event K1 Count:	Number of times cup sensor calibration failed due the Top cup sensor being covered by mess	3

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Menu Title of Counter	Description	Severity
Event L1 Count:	Number of times an unwanted pack was cleared away before the start of a "no pack" vend	0
Event T1 Count:	Number of times the pack door opening motor cam overran its 'Home' position at the end of a door opening cycle (see Note 6)	6
Event U1 Count:	Number of times the drink selection process was aborted by the pack door unlatching and the pack disappearing	6
Event V1 Count:	Number of times one of the hot tanks failed to heat up	9
Event W1 Count:	Number of times the Pre hot tank overshot its target temperature	5
Event X1 Count:	Number of times the Main hot tank overshot its target temperature	5

Note 1: the system pressure switch cuts power to both the Air pump and the Water pump if it senses excessive pressure in the pipe run to the pack injector. The control system can detect if the switch has activated and will immediately turn off the pump for a period of time (1.4 secs) to give time for the pressure to subside. It will then attempt to start the pump again and continue with that section of the brew cycle. If this retry process occurs more than 8 times in any section of a brew cycle then **PCO 5** will be logged (and the vend will be aborted via a special de-watering process); if the retry process succeeds then **Event I** will be logged (and the vend will continue). These two counters are specifically for errors/events triggered by consumers vending drinks; they deliberately **exclude** errors/events triggered by engineering menu commands.

Note 2: this event is detected 0.4 secs after the barrel cam and injector reaches the top home position at the end of a pack ejection cycle. It is detected at the end of *all* ejection cycles.

Note 3: this event is logged when the user confirms on the consumer interface that they emptied the internal pack bin before they replaced it.

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Note 4: these counters are specifically for errors/events triggered by consumers selecting a drink in vending mode; they deliberately **exclude** errors/events triggered in the engineering menus.

Note 5: the "pack absent" timeout aborts a vend (but does not cause a PCO error) if a pack is not detected for 3.5 seconds during a vend when the pack door is closed.

Note 6: this event is detected 0.2 secs after the cam reaches its home position at the end of a pack door opening cycle. It is detected at the end of *all* door opening cycles.

Note 7: "Around pack injection time" means from the moment that the handling mech motor is started to the end of the brew cycle segment that immediately follows the injection. These events normally indicate that the injector and the filter pack are not aligned properly; this can lead to more serious faults such as "pack inject timeouts" (see '[PCO 1](#)') and "pack absent" timeouts (see '[Event F](#)' and Note 5).

Note 8: the pack detect switch should normally precede the pack door switch when the pack door is shut to start a vend; in case this does not occur, the software waits for up to 0.4 secs for a pack to appear after the pack door is closed. If a pack *does* appear during this period then this event is logged and the vend is started; if a pack *is not* detected then the drink selection is cleared and the consumer interface returns to the 'select drink' state.

Note 9: the "pawl arm reset cycle" is carried out by moving the barrel cam and injector through one complete cycle (i.e. from top to bottom to top again). It is triggered by the conditions for '[Event H](#)' (ref. Note 10). If the pawl arm is still not home at the end of the reset cycle then this event is logged.

Note 10: this event is detected 0.4 secs after the barrel cam and injector reaches its top home position at the end of a pack ejection cycle. This event is detected at the end of all pack ejection cycles except those initiated by the engineering menus.

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Note 11: the “internal watchdog” is a hardware module within the main control board’s H8 microprocessor that will internally reset the micro if it is not updated regularly (every 213ms or less) by the software. The “external watchdog” is an IC device on the main control board that will externally reset the H8 micro if it is not updated regularly (every 1 sec) by a software-driven output from the micro; this watchdog comes in to play if the H8 micro completely locks up (e.g. because of an electrostatic discharge).

Note 12: to detect problems with the Barcode scan engine, a certain non-default configuration setting is monitored periodically; if its value ever changes (e.g. scan engine memory corrupted or scan engine controller re-booted due to a crash or a poor connection) then all the initialisation commands are re-sent to the Barcode scan engine and ‘[Event Q](#)’ is logged.

Note 13: in Vend Mode, the barcode scanner is triggered when the pack door is closed with a pack present. If no barcode data is received within 0.45 secs then ‘[Event T](#)’ is logged and the consumer interface proceeds without pack identification. As well as total failures to read a barcode, it is also possible for the reading to be so slow that it exceeds the timeout. The barcode engineering menus have a much longer or no timeout and can read poor quality or poorly aligned barcodes that can’t be read in Vend Mode.

Note 14:

Note 15: this event is detected when the main power supply falls below a certain level but then recovers rather than dying away altogether. However, if the duration of the power dip is too great then the subsequent boot-up will be classed as a “power-up” event and counted separately (see ‘[Event K](#)’). Brown-out events indicate either a poor quality mains power supply to the brewer or faulty wiring inside or outside of the brewer (but note that they can also be caused by flicking the power switch off/on at the right rate).

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Note 16: if the cup detection system has been enabled in the menus then its signals will be monitored during each vend. If a cup is removed *during a vend* then, depending on the stage the brew cycle has reached, this may cause the brew cycle either to pause (ultimately to timeout and abort) or to immediately abort. Any such aborts are logged in [Events U](#) and [V](#) as appropriate. If a cup is removed during the final stages of de-watering then the brew cycle will *not* be paused or aborted and no [Event](#) will be logged.

Note 17: if a cup is removed during the main, *non-jet-water* phases of a vend then the brew cycle will be paused until the cup is replaced or has been absent for 6 seconds. If the cup is not replaced within 6 seconds then [‘Event U’](#) will be logged and the brew cycle will be terminated as quickly and as cleanly as possible (i.e. the pack will be de-watered if necessary and then ejected).

Note 18: the “jet water phase” of a brew cycle means from the start of pumping water towards the jet nozzle to the point just after water finishes coming out of the jet. If a cup is removed during this phase then the brew cycle will *immediately* be terminated as quickly and as cleanly as possible (de-watering will occur if necessary) and [‘Event V’](#) will be logged. Note that there is no pause and no opportunity to resume the brew cycle by replacing the cup.

Note 19:

Note 20: if the pack fan speed is measured at consistently less than 206Hz for more than 4 seconds then the fan is deemed to be running slow and [PCO 14](#) will be logged.

Note 21: if the pack fan is detected as stationary when it is enabled it will be given up to 3 subsequent attempts to start moving. If none of these retries successfully start the pack fan then the fan is deemed to have [stalled](#) and [PCO 13](#) will be logged. If one of the retries successfully starts the pack fan then [‘Event Y’](#) will be logged.

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